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Communications Systems with Header and Acknowledgement Integrity Protection

Abstract

A communication system is provided in which access points (APs) communicate with stations (STAs). The APs and STAs may convey integrity-protected frames. The APs and STAs may integrity check the integrity-protected frames. The APs and STAs may transmit integrity-protected block acknowledgement (BA) frames in response to receiving integrity-protected frames. The APs and STAs may integrity check the integrity-protected BA frames. Padding may be inserted into the integrity-protected frames and/or BA frames to accommodate processing delays associated with generating and verifying message integrity checks (MICs) in the frames and/or BA frames. Integrity checking the frames may be performed before, after, or both before and after transmitting a corresponding BA frame.

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Background/Summary

[0001] This application claims the benefit of U.S. Provisional Patent Application No. 63/561,087, filed Mar. 4, 2024, which is hereby incorporated by reference herein in its entirety.

FIELD

[0002] This disclosure relates generally to wireless communications, including wireless communications by electronic devices.

BACKGROUND

[0003] Communications systems and methods are used to convey wireless data between nodes of a communications network. The nodes can include user equipment devices, wireless access points, wireless base stations, or other electronic devices.

[0004] It can be challenging to ensure that communications systems exhibit sufficient levels of performance. If care is not taken, wireless data transmitted by a first device might not be properly received at a second device. In addition, the wireless data can be susceptible to security risks such as man-in-the-middle (MITM) attacks while being conveyed between the first and second devices.

SUMMARY

[0005] A communication system is provided in which access points (APs) communicate with stations (STAs). The APs and STAs may convey integrity-protected frames. The APs and STAs may integrity check the integrity-protected frames. The APs and STAs may transmit integrity-protected block acknowledgement (BA) frames in response to receiving integrity-protected frames. The APs and STAs may integrity check the integrity-protected BA frames. Padding may be inserted into the integrity-protected frames and/or BA frames to accommodate processing delays associated with generating and verifying message integrity checks (MICs) in the frames and/or BA frames. Integrity checking the frames may be performed before, after, or both before and after transmitting a corresponding BA frame.

[0006] An aspect of the disclosure provides a method of operating a first electronic device to wirelessly communicate with a second electronic device. The method can include transmitting, using one or more antennas, an integrity-protected frame for receipt by the second electronic device. The integrity-protected frame can include a preamble, a media access control (MAC) header, a delimiter between the preamble and the MAC header, a message integrity check (MIC) field based on the MAC header, a data payload, the MAC header being between the data payload and the delimiter, and a padding field configured to accommodate a processing delay associated with generating or verifying the MIC field.

[0007] An aspect of the disclosure provides a method of operating a first electronic device to wirelessly communicate with a second electronic device. The method can include receiving, using one or more antennas, a frame. The method can include transmitting, using the one or more antennas, an integrity-protected block acknowledgement (BA) frame for receipt by the second electronic device. The integrity-protected BA frame can include a preamble, a block acknowledgment (ACK) to a media access control protocol data unit (MPDU) in the frame, a message integrity check (MIC) field based on the block ACK, the block ACK being between the MIC field and the preamble in the integrity-protected BA frame, and a padding field configured to accommodate a processing delay associated with generating or verifying the MIC field.

[0008] An aspect of the disclosure provides a method of operating a first electronic device to wirelessly communicate with a second electronic device. The method can include receiving, using

one or more antennas, a frame. The method can include transmitting, using the one or more antennas, a block acknowledgement (BA) frame for receipt by the second electronic device, wherein the BA frame includes a block acknowledgment (ACK) to a media access control protocol data unit (MPDU) in the frame. The method can include attempting to verify, using one or more processors prior to transmission of the integrity-protected BA frame, integrity of a media access control (MAC) header of the MPDU in the frame.

[0009] An aspect of the disclosure provides a method of operating a first electronic device to wirelessly communicate with a second electronic device. The method can include receiving, using one or more antennas, a frame. The method can include transmitting, using the one or more antennas, a block acknowledgement (BA) frame for receipt by the second electronic device, wherein the BA frame includes a block acknowledgment (ACK) to a media access control protocol data unit (MPDU) in the frame. The method can include calculating, using one or more processors after of the integrity-protected BA frame, a message integrity check (MIC) value based on at least some of the MPDU. The method can include transmitting, using the one or more antennas, a block acknowledgment request (BAR) frame when the calculated MIC value does not match a corresponding MIC field included in the frame, wherein the BAR frame identifies the MPDU.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1 is a diagram of an illustrative wireless communications system in with some embodiments.

[0011] FIG. 2 is a schematic diagram of an illustrative wireless station (STA) in accordance with some embodiments.

[0012] FIG. 3 is a schematic diagram of an illustrative wireless access point (AP) in accordance with some embodiments.

[0013] FIG. 4 is a diagram of an illustrative wireless communications system including a first electronic device that communicates with a second electronic device over a wireless communication link in accordance with some embodiments.

[0014] FIG. 5 is a timing diagram of an illustrative integrity-protected frame and a corresponding integrity-protected block acknowledgement (BA) frame that may be conveyed between first and second electronic devices in accordance with some embodiments.

[0015] FIG. 6 is a flow chart of illustrative operations involved in integrity protecting and integrity checking wireless data conveyed between first and second electronic devices in accordance with some embodiments.

[0016] FIGS. 7A-7D are timing diagrams showing how an illustrative integrity-protected frame may be provided with padding at different locations in accordance with some embodiments.

[0017] FIG. 8 is a timing diagram showing how an illustrative integrity-protected physical protocol data unit (PPDU) frame may be provided with padding between different media access control protocol data units (MPDUs) in accordance with some embodiments.

[0018] FIGS. 9A and 9B are timing diagrams showing how an illustrative integrity-protected frame may be provided with padding at different locations in accordance with some embodiments.

[0019] FIG. 10 is a flow chart of illustrative operations involved in using a receiver to integrity check a received frame after transmitting a corresponding BA frame in accordance with some embodiments.

[0020] FIG. 11 is a flow chart of illustrative operations involved in using a receiver to integrity check a received frame both before and after transmitting a corresponding BA frame in accordance with some embodiments.

[0021] FIG. 12 is a flow chart of illustrative operations involved in using a receiver to integrity

check a received frame prior to transmitting a corresponding BA frame in accordance with some embodiments.

[0022] FIG. **13** is a diagram showing an illustrative signaling scheme between a multi-layer multi-device (MLMD) AP and an MLMD STA in accordance with some embodiments.

[0023] FIG. **14** is a diagram showing how a header message integrity check (MIC) may be placed at different locations within an illustrative integrity-protected frame in accordance with some embodiments.

[0024] FIG. **15** is a diagram showing how an illustrative integrity-protected frame may include a combined MIC field that integrity protects both a header and a data payload of the integrity-protected frame in accordance with some embodiments.

[0025] FIG. **16** is a diagram of an illustrative integrity-protected multi-STA BA frame having a configurable AID11 field in accordance with some embodiments.

[0026] FIG. **17** shows a table outlining different illustrative AID11 fields that may be included in an integrity-protected multi-STA BA frame in accordance with some embodiments.

[0027] FIG. **18** is a diagram of two illustrative integrity-protected frames that include time synchronization function (TSF) information for soliciting a corresponding BA frame in accordance with some embodiments.

[0028] FIG. **19** is a diagram showing how data frames and BA frames may be conveyed between an illustrative multi-link device (MLD) transmitter and an illustrative MLD receiver in accordance with some embodiments.

[0029] FIG. **20** is a timing diagram showing how illustrative first and second devices may mitigate BA off-synchronization using a silent reorder buffer update at the second device in accordance with some embodiments.

[0030] FIG. **21** is a timing diagram showing how illustrative first and second devices may mitigate BA off-synchronization using a transmission buffer update delay at the first device in accordance with some embodiments.

[0031] FIG. **22** is a diagram showing how reorder buffer status information may be included in the BA control field of an illustrative integrity-protected multi-STA BA frame in accordance with some embodiments.

[0032] FIG. **23** shows a table outlining different illustrative AID11 fields that may be included in an integrity-protected multi-STA BA frame having a BA control field with reorder buffer status information in accordance with some embodiments.

DETAILED DESCRIPTION

[0033] FIG. **1** illustrates an example of a wireless communication system **108** (sometimes also referred to herein as wireless communications network **108**, communications network **108**, network **108**, or system **108**). It is noted that FIG. **1** represents one possibility among many, and that features of the present disclosure may be implemented in any of various systems, as desired. For example, embodiments described herein may be implemented in any type of wireless device. The wireless embodiment described below is one example embodiment.

[0034] As shown in FIG. **1**, the exemplary wireless communication system **108** includes an access point (AP) **104**, which communicates over a transmission medium with one or more wireless devices **106** (e.g., a first wireless device **106A**, a second wireless device **106B**, etc.). Wireless devices **106A** and **106B** may be user devices (e.g., user equipment (UE) devices), such as stations (STAs), non-AP STAs, or wireless local area network (WLAN) devices. Wireless devices **106** are sometimes referred to herein as STAs **106**.

[0035] STA **106** may be a device with wireless network connectivity such as a mobile (e.g., cellular) telephone, a hand-held device, a wearable device (e.g., a wristwatch device, pendant device, ring device, head-mounted device such as a virtual, mixed, and/or augmented reality headset, goggles, helmet, or glasses, etc.), a computer (e.g., a desktop computer, laptop computer, a computer monitor containing an embedded computer, etc.), a tablet computer, a media player,

headphones, one or two wireless earbuds, a television, a gaming device or console, a navigation device, an embedded system such as a system in which electronic equipment with a display is mounted in a kiosk or automobile, a wireless internet-connected voice-controlled speaker, a home entertainment device, a remote control device, a gaming controller, a user input device, peripheral, or accessory, an electronic stylus or pen, an unmanned aerial vehicle (UAV), an unmanned aerial controller (UAC), an automobile, computing equipment integrated into a vehicle or kiosk, equipment that implements the functionality of two or more of these devices, or virtually any type of wireless device.

[0036] STA **106** may include a processor (processing element) that is configured to execute program instructions stored in memory. STA **106** may perform any of the method embodiments described herein by executing such stored instructions. Alternatively, or in addition, STA **106** may include a programmable hardware element such as an FPGA (field-programmable gate array), an integrated circuit, and/or any of various other possible hardware components that are configured to perform (e.g., individually or in combination) any of the method embodiments described herein, or any portion of any of the method embodiments described herein.

[0037] Wireless communications system **108** may include one or more wireless access points (APs) such as AP **102**. AP **102** may be a stand-alone AP or an enterprise AP and may include hardware that enables wireless communication with STAs **106** such as STA **106A** and STA **106B**. AP **102** may also be equipped to communicate with a network **100** (e.g., a WLAN, an enterprise network, and/or another communication network connected to the Internet, among various possibilities). Thus, AP **102** may facilitate communication among STA devices **106** and/or between STA devices **106** and network **100**. AP **102** can be configured to provide communications over one or more wireless technologies, such as any of 802.11 a, b, g, n, ac, ad, ax, ay, be, bn, and/or other 802.11 versions, or a cellular protocol, such as 5G or LTE, including in an unlicensed band (LAA).

[0038] Network **100** may include any desired number of network nodes, terminals, and/or end hosts that are communicably coupled together using communications paths that include wired and/or wireless links. The wired links may include cables (e.g., ethernet cables, optical fibers or other optical cables that convey signals using light, telephone cables, radio-frequency cables such as coaxial cables or other transmission lines, etc.). The wireless links may include short range wireless communications links that operate over a range of inches, feet, or tens of feet, medium range wireless communications links that operate over a range of hundreds of feet, thousands of feet, miles, or tens of miles, and/or long range wireless communications links that operate over a range of hundreds or thousands of miles.

[0039] The nodes of network **100** may be organized into one or more relay networks, mesh networks, local area networks (LANs), wireless local area networks (WLANs), ring networks (e.g., optical rings), cloud networks, virtual/logical networks, the Internet (e.g., may be communicably coupled to each other over the Internet), combinations of these, and/or using any other desired network topologies. The network nodes, terminals, and/or end hosts of network **100** may include network switches, network routers, optical add-drop multiplexers, other multiplexers, repeaters, modems, portals, gateways, servers, network cards (line cards), wireless access points, wireless base stations, and/or any other desired network components. The network nodes in network **100** may include physical components such as electronic devices, servers, computers, network racks, line cards, user equipment, etc., and/or may include virtual components that are logically defined in software and that are distributed across (over) two or more underlying physical devices (e.g., in a cloud network configuration).

[0040] The communication area (or coverage area) of AP **102** (or AP **104**) may be referred to as a basic service area (BSA) or cell. AP **102** (or AP **104**) and STAs **106** may be configured to communicate over the transmission medium using any of various radio access technologies (RATs) or wireless communication technologies, such as Wi-Fi, LTE, LTE-Advanced (LTE-A), 5G NR, ultra-wideband (UWB), etc. A given RAT may, for example, specify the physical methodology used

in implementing a corresponding communications protocol (e.g., a WLAN protocol, a wireless personal area network (WPAN) protocol, a cellular telephone protocol such as a 3G protocol, a 4G (LTE) protocol, a 5G (NR) protocol, etc., a UWB protocol, a satellite communications protocol, a satellite navigation protocol, a device-to-device (D2D) protocol, etc.).

[0041] AP **102**, AP **104**, and other similar access points (not shown) operating according to one or more wireless communication technologies may thus be provided as a network, which may provide continuous or nearly continuous overlapping service to STAs **106A** and **106B** and similar devices over a geographic area (e.g., via one or more communication technologies). A STA may roam from one AP to another AP directly or may transition between APs and cellular network cells, for example.

[0042] Note that at least in some instances STA **106** may be capable of communicating using any of multiple wireless communication technologies. For example, STA **106** might be configured to communicate using one or more of Wi-Fi, LTE, LTE-A, 5G NR, Bluetooth, UWB, one or more satellite systems, etc. Other combinations of wireless communication technologies (including more than two wireless communication technologies) are also possible. Likewise, in some instances STA **106** can be configured to communicate using only a single wireless communication technology.

[0043] As shown in FIG. **1**, the exemplary wireless communication system **108** can also include an AP **104**, which communicates over a transmission medium with the wireless device **106B**. AP **104** also provides communicative connectivity to network **100**. Thus, according to some embodiments, wireless devices may be able to connect to either or both of AP **102** (or a cellular base station (BS)) and AP **104** (or another access point) to access the network **100**. For example, a STA may roam from AP **102** to AP **104** based on one or more factors, such as coverage, interference, and capabilities. Note that it may also be possible for AP **104** to provide access to a different network (e.g., an enterprise Wi-Fi network, a home Wi-Fi network, etc.) than the network to which the AP **102** provides access.

[0044] In some implementations, STAs **106** (e.g., STAs **106A** and **106B**) may include handheld devices such as smart phones or tablets, wearable devices such as smart watches or smart glasses, and/or may include any of various types of devices with wireless communication capability. For example, one or more of the STAs **106A** and/or **106B** may be a wireless device intended for stationary or nomadic deployment such as an appliance, measurement device, control device, etc.

[0045] STA **106B** may also be configured to communicate with STA **106A**. For example, STA **106A** and STA **106B** may be capable of performing direct device-to-device (D2D) communication. In some embodiments, such direct communication between STAs may also or alternatively be referred to as peer-to-peer (P2P) communication. The direct communication may be supported by AP **102** (e.g., AP **102** may facilitate discovery, among various possible forms of assistance), or may be performed in a manner unsupported by the AP **102**. Such P2P communication may be performed using 3GPP-based D2D communication techniques, Wi-Fi-based P2P communication techniques, UWB, Bluetooth (BT), and/or any of various other direct communication techniques, according to various embodiments.

[0046] STA **106** may include one or more devices or integrated circuits for facilitating wireless communication, potentially including a WLAN (e.g., Wi-Fi) modem, a cellular modem, and/or one or more other wireless modems. The wireless modem(s) may include one or more processors (processor elements) and various hardware components as described herein. STA **106** may perform any of (or any portion of) the method embodiments described herein by executing instructions on one or more programmable processors. Alternatively, or in addition, the one or more processors may be one or more programmable hardware elements such as an FPGA (field-programmable gate array), or other circuitry, that is configured to perform any of the method embodiments described herein, or any portion of any of the method embodiments described herein. The wireless modem(s) described herein may be used in a STA as defined herein, a wireless device as defined herein, or a communication device as defined herein. The wireless modem described herein may also be used

in an AP, a base station, a pico cell, a femto cell, or other similar network side device.

[0047] STA **106** may include one or more antennas for communicating using one or more wireless communication protocols or radio access technologies. In some embodiments, STA **106** can be configured to communicate using a single shared radio. The shared radio may couple to a single antenna, or may couple to multiple antennas (e.g., for multiple-input-and-multiple-output (MIMO)) for performing wireless communications. Alternatively, STA **106** may include two or more radios, each of which may be configured to communicate via a respective wireless link. Other configurations are also possible.

[0048] FIG. 2 is one possible block diagram of a STA device such as STA **106**. STA **106** is sometimes also referred to herein as UE **106**, UE device **106**, or device **106**. STA **106** also may be referred to herein as non-AP STA **106** or non-AP device **106**. As shown in FIG. 2, STA **106** may include wireless circuitry such as wireless communication circuitry **230**, a subsystem such as system on chip (SOC) **200**, a display such as display **260**, and one or more interfaces such as connector interface (I/F) **220**.

[0049] SOC **200** may include one or more portions configured for various purposes. For example, as shown in FIG. 2, SOC **200** may include one or more processors **202** and display circuitry **204**. Processor(s) **202** may execute program instructions for STA **106**. Display circuitry **204** may perform graphics processing and may provide display signals to display **260**. Display **260** may be a touch-sensitive display, a force sensitive display, or a display without touch or force sensitivity. Display **260** may include one or more arrays of display pixels that emit light containing images, for example.

[0050] SOC **200** may also include sensor circuitry such as motion sensing circuitry **270**. Motion sensing circuitry **270** may detect motion of the STA **106** using, for example, a gyroscope, accelerometer, inertial measurement unit (IMU), compass, and/or any of various other motion sensing components. Processor(s) **202** may also be coupled to memory management unit (MMU) **240**, which may be configured to receive addresses from processor(s) **202** and may translate those addresses to locations in memory or other storage circuitry (e.g., memory **206**, read only memory (ROM) **250**, flash (NAND) memory **210**, etc.). MMU **240** may be configured to perform memory protection and page table translation or set up. In some embodiments, MMU **240** may be included as a portion of processor(s) **202**.

[0051] SOC **200** may be coupled to various other circuits in STA **106**. For example, SOC **200** may be coupled to various types of memory (e.g., flash memory **210**), connector interface **220** (e.g., for coupling to a computer system, dock, charging station, etc.), display **260**, and wireless communication circuitry **230** (e.g., for performing wireless communications under LTE, LTE-A, 5G NR, Bluetooth, Wi-Fi, NFC, GPS, UWB, etc.).

[0052] STA **106** may include at least one antenna **235**. If desired, STA **106** may include multiple antennas **235** such as at least a first antenna **235A** and a second antenna **235B**. STA **106** may use antennas **235** to perform wireless communication with access points, base stations, and/or other devices. For example, STA **106** may use antennas **235A** and **235B** to perform the wireless communication with APs **102** and/or **104** of FIG. 1. As noted above, STA **106** may, in some embodiments, be configured to communicate wirelessly using multiple wireless communication standards or radio access technologies (RATs).

[0053] Wireless communication circuitry **230** may include one or more modems such as WLAN (e.g., Wi-Fi) modem **232**, cellular modem **234**, and Bluetooth modem **236**. If desired, wireless communication circuitry **230** may include additional modems for handling other RATs or wireless communications technologies. STA **106** may use WLAN modem **232** (sometimes also referred to herein as Wi-Fi modem **232**) to perform Wi-Fi or other WLAN communications (e.g., on an 802.11 network) with one or more external devices (e.g., AP **104** and/or **102** of FIG. 1). STA **106** may use Bluetooth modem **236** to perform Bluetooth communications or other WPAN communications with one or more external devices (e.g., another STA **106**). STA **106** may use cellular modem **234** to

perform cellular communications with one or more wireless base stations according to one or more cellular communication technologies (e.g., in accordance with one or more 3GPP specifications). [0054] As described herein, STA **106** may include hardware and software components for implementing embodiments of this disclosure. For example, one or more components of the wireless communication circuitry **230** (e.g., Wi-Fi modem **232**, cellular modem **234**, BT modem **236**) of the STA **106** may be configured to implement part or all of the methods described herein, e.g., by one or more processors executing program instructions stored on a memory medium (e.g., a non-transitory computer-readable memory medium), a processor configured as an FPGA (Field Programmable Gate Array), and/or using dedicated hardware components, which may include an ASIC (Application Specific Integrated Circuit). STA **106** may include support structures such as a housing. The housing may include conductive and/or dielectric housing walls, layers, and/or other structures.

[0055] If desired, STA **106** may include additional include input-output devices (not shown for the sake of clarity). The input-output devices may be used to allow data to be supplied to STA **106** and to allow data to be provided from STA **106** to external devices. The input-output devices may include user interface devices, data port devices (e.g., interface **220**), touch sensors, displays (e.g., display **260**), light-emitting components such as displays without touch sensor capabilities, buttons (mechanical, capacitive, optical, etc.), scrolling wheels, touch pads, key pads, keyboards, microphones, cameras, buttons, speakers, status indicators, audio jacks and other audio port components, digital data port devices, motion sensors (accelerometers, gyroscopes, and/or compasses that detect motion), capacitance sensors, proximity sensors, magnetic sensors, force sensors (e.g., force sensors coupled to a display to detect pressure applied to the display), temperature sensors, etc. In some configurations, keyboards, headphones, displays, pointing devices such as trackpads, mice, and joysticks, and other input-output devices may be coupled to STA **106** using wired or wireless connections (e.g., some of the input-output devices may be peripherals that are coupled to a main processing unit or other portion of STA **106** via a wired or wireless link).

[0056] FIG. **3** is an example block diagram of an electronic device such as AP **104** (or equivalently AP **102** of FIG. **1**). In some instances (e.g., in an 802.11 communication context), AP **104** may also be referred to as a station (STA), and possibly more particularly as an AP STA. It is noted that the AP of FIG. **3** is merely one example of a possible access point. As shown, AP **104** may include one or more processors **304**, which may execute program instructions for AP **104**. Processor(s) **304** may also be coupled to MMU **340**, which may be configured to receive addresses from processor(s) **304** and to translate those addresses to locations in memory (e.g., memory **360** and ROM **350**) or to other storage circuitry, circuits, or devices.

[0057] AP **104** may include at least one network port **370**. Network port **370** may be configured to couple to a network and to provide multiple devices, such as STAs **106**, with access to the network (e.g., network **100** of FIG. **1**). Network port **370** (or an additional network port) may also or alternatively be configured to couple to a cellular network (e.g., a core network (CN) of a cellular service provider). The core network may provide mobility related services and/or other services to a plurality of UE devices (e.g., STAs **106**). In some cases, network port **370** may couple to a telephone network via the core network, and/or the core network may provide a telephone network (e.g., among other UE devices serviced by the cellular service provider).

[0058] AP **104** may include one or more radios **330A-330N**, each of which may be coupled to a respective communication chain **332** and at least one antenna **334**, and possibly multiple antennas (e.g., a first radio **330A** coupled to antenna **334A** via communication chain **332A**, an Nth radio **330N** coupled to antenna **334N** via communication chain **332N**, etc.). Radios **330** may be configured to operate as wireless transceivers that communicate with STAs **106** via communication chains **332** and antennas **334**. Antenna(s) **334A-N** communicate with their respective radios **330A-N** via communication chains **332A-N**. Communication chains **332** may be receive chains, may be

transmit chains, or may include both transmit and receive chains. Radios **330A-N** may be configured to communicate in accordance with various wireless communication standards including, but not limited to, LTE, LTE-A, 5G NR, 6G, UWB, WLAN (Wi-Fi), WPAN (BT), etc. If desired, AP **104** may be configured to operate on multiple wireless links using the one or more radios **330A-N**, where each radio is used to operate on a respective wireless link.

[0059] AP **104** may be configured to communicate wirelessly using one or multiple wireless communication standards. In some instances, AP **104** may include multiple radios, which may enable the network entity to communicate according to multiple wireless communication technologies. For example, as one possibility, AP **104** may include an LTE or 5G NR radio for performing communication according to LTE or 5G as well as a Wi-Fi radio for performing communication according to Wi-Fi. In such a case, AP **104** may be capable of operating as both a cellular base station and a Wi-Fi access point. As another possibility, AP **104** may include a multi-mode radio, which is capable of performing communications according to any of multiple wireless communication technologies (e.g., NR and Wi-Fi, NR and LTE, etc.). As still another possibility, AP **104** may be configured to act exclusively as a Wi-Fi access point, e.g., without cellular communication capability.

[0060] As described further herein, AP **104** may include hardware and software components for implementing or supporting implementation of features described herein. Processor(s) **304** of AP **104** may be configured to implement, or support implementation of, part or all of the methods described herein, e.g., by executing program instructions stored on a memory medium (e.g., a non-transitory computer-readable memory medium) to operate multiple wireless links using multiple respective radios. Alternatively, processor(s) **304** may be configured as a programmable hardware element, such as an FPGA (Field Programmable Gate Array) or ASIC (Application Specific Integrated Circuit), or a combination thereof. Alternatively (or in addition) the processor(s) **304** of AP **104**, in conjunction with one or more of the other components **330, 332, 334, 340, 350, 360, 370** may be configured to implement, or support implementation of, part or all of the features described herein.

[0061] Radio(s) **330** on AP **104** may use antenna(s) **334** (FIG. 3) and wireless communication circuitry **230** on STA **106** may use antenna(s) **235** (FIG. 2) to transmit and/or receive radio-frequency signals within different frequency bands at radio frequencies (sometimes referred to herein as communications bands or simply as a “bands”). The frequency bands handled by AP **104** and STA **106** may include satellite communications bands (e.g., the C band, S band, L band, X band, W band, V band, K band, K.sub.a band, K.sub.u band, etc.), wireless local area network (WLAN) frequency bands (e.g., Wi-Fi® (IEEE 802.11) or other WLAN communications bands) such as a 2.4 GHz WLAN band (e.g., from 2400 to 2480 MHz), a 5 GHz WLAN band (e.g., from 5180 to 5825 MHz), a Wi-Fi® 6E band (e.g., from 5925-7125 MHz), and/or other Wi-Fi® bands (e.g., from 1875-5160 MHz), wireless personal area network (WPAN) frequency bands such as the 2.4 GHz Bluetooth® band or other WPAN communications bands, cellular telephone frequency bands (e.g., bands from about 600 MHz to about 5 GHz, 3G bands, 4G LTE bands, 5G New Radio Frequency Range 1 (FR1) bands below 10 GHz, 5G New Radio Frequency Range 2 (FR2) bands between 20 and 60 GHz, 6G bands, etc.), other centimeter or millimeter wave frequency bands between 10-300 GHz, near-field communications (NFC) frequency bands (e.g., at 13.56 MHz), satellite navigation frequency bands (e.g., a GPS band from 1565 to 1610 MHz, a Global Navigation Satellite System (GLONASS) band, a BeiDou Navigation Satellite System (BDS) band, etc.), ultra-wideband (UWB) frequency bands that operate under the IEEE 802.15.4 protocol and/or other ultra-wideband communications protocols, communications bands under the family of 3GPP wireless communications standards, communications bands under the IEEE 802.XX family of standards, and/or any other desired frequency bands of interest.

[0062] Antenna(s) **334** (FIG. 3) and antenna(s) **235** (FIG. 2) may be formed using any desired antenna structures. For example, the antennas may include antennas with resonating elements that

are formed from loop antenna structures, patch antenna structures, inverted-F antenna structures, slot antenna structures, planar inverted-F antenna structures, helical antenna structures, monopole antennas, dipoles, hybrids of these designs, etc. If desired, one or more antennas may include antenna resonating elements formed from conductive portions of a device housing (e.g., peripheral conductive housing structures extending around a periphery of a display on STA **106**). Filter circuitry, switching circuitry, impedance matching circuitry, and/or other antenna tuning components may be adjusted to adjust the frequency response and wireless performance of the antennas over time. If desired, multiple antennas may be implemented as a phased array antenna (e.g., where each antenna forms a radiator or antenna element of the phased array antenna, which is sometimes also referred to as a phased antenna array). In these scenarios, the phased array antenna may convey radio-frequency signals within a signal beam. The phases and/or magnitudes of each radiator in the phased array antenna may be adjusted so the radio-frequency signals for each radiator constructively and destructively interfere to steer or orient the signal beam in a particular pointing direction (e.g., a direction of peak signal gain). The signal beam may be adjusted or steered over time.

[0063] Wireless communication circuitry **230** may convey radio-frequency signals using antenna(s) **235** (FIG. 2). Radio(s) **330** may convey radio-frequency signals using antenna(s) **334** (FIG. 3). The term “convey radio-frequency signals” as used herein means the transmission and/or reception of the radio-frequency signals (e.g., for performing unidirectional and/or bidirectional wireless communications with external wireless communications equipment). The term “convey wireless data” as used herein means the transmission and/or reception of the wireless data (e.g., as carried by corresponding radio-frequency signals). Antennas may transmit radio-frequency signals by radiating the radio-frequency signals into free space (or to free space through intervening device structures such as a dielectric cover layer). Antennas may additionally or alternatively receive radio-frequency signals from free space (or through intervening devices structures such as a dielectric cover layer). The transmission and reception of radio-frequency signals by antennas each involve the excitation or resonance of antenna currents on an antenna resonating element in the antenna by the radio-frequency signals within the frequency band(s) of operation of the antenna.

[0064] Wireless communication circuitry **230** may be coupled to antenna(s) **235** (FIG. 2) over one or more radio-frequency transmission lines. Radio(s) **330** may be coupled to antenna(s) **334** (FIG. 3) over one or more radio-frequency transmission lines. Communication chain(s) **332** (FIG. 3) may be disposed on the radio-frequency transmission lines between antenna(s) **334** and radio(s) **330**. The radio-frequency transmission lines may include coaxial cables, microstrip transmission lines, stripline transmission lines, edge-coupled microstrip transmission lines, edge-coupled stripline transmission lines, transmission lines formed from combinations of transmission lines of these types, etc. The radio-frequency transmission lines may be integrated into rigid and/or flexible printed circuit boards if desired. One or more of the radio-frequency lines may be shared between radios or modems if desired. Radio-frequency front end (RFFE) modules may be interposed on one or more of the radio-frequency transmission lines if desired (e.g., within communication chain(s) **332** of FIG. 3 or within wireless communication circuitry **230** of FIG. 2). The radio-frequency front end modules may include substrates, integrated circuits, chips, or packages that are separate from the radios or modems and may include filter circuitry, switching circuitry, amplifier circuitry, impedance matching circuitry, radio-frequency coupler circuitry, and/or any other desired radio-frequency circuitry for operating on the radio-frequency signals conveyed over the radio-frequency transmission lines.

[0065] Processor(s) **202** (FIG. 2) and processor(s) **304** (FIG. 3) may each include one or more processors such as microprocessors, microcontrollers, digital signal processors, host processors, baseband processing circuitry (e.g., one or more baseband processors or baseband processor integrated circuits), application specific integrated circuits (ASICs), FPGAs, central processing units (CPUs), graphics processing units (GPUs), etc. If desired, radio(s) **330** (FIG. 3) and/or

wireless communication circuitry **230** may also include one or more processors. Baseband circuitry in STA **106** and/or AP **104** may, for example, access a communication protocol stack on corresponding storage circuitry (e.g., memory **206** of FIG. 2 or memory **360** of FIG. 3) to: perform user plane functions at a PHY layer, MAC layer, RLC layer, PDCP layer, SDAP layer, and/or PDU layer, and/or to perform control plane functions at the PHY layer, MAC layer, RLC layer, PDCP layer, RRC, layer, and/or non-access stratum layer.

[0066] AP **104** (or AP **102** of FIG. 1) may communicate with a STA **106** over a corresponding wireless communication link. Radio-frequency signals may be wirelessly conveyed between the radios and antennas on AP **104** and STA **106** to support the wireless communication link. The radio-frequency signals may include wireless data modulated onto one or more carriers of the radio-frequency signal (e.g., by a transmitter in radio **330** of AP **104** or a transmitter in a modem on wireless communication circuitry **230** of STA **106**). The wireless data may be organized, modulated onto the radio-frequency signals, and demodulated from the radio-frequency signals (e.g., by a receiver in radio **330** of AP **104** or a receiver in a modem on wireless communication circuitry **230** of STA **106**) according to a corresponding communications protocol or standard (e.g., an IEEE 802.11 protocol or standard). The radio-frequency signals may be conveyed in one or more frequency bands associated with the communications protocol.

[0067] Implementations in which AP **104** and STA **106** communicate according to an IEEE 802.11 protocol or standard are described herein as an example. Under an 802.11 protocol, the wireless data is organized into a series or flow of frames (e.g., media access control (MAC) frames) carried by the radio-frequency signals. The frames, which are sometimes also referred to as packets, may include management frames, control frames, data frames, beacon frames, association frames, authentication frames, acknowledgement (ACK) frames, block ACK frames, and/or other types of frames. Each frame may include a frame header, body (e.g., after the header), and trailer (e.g., after the body). The header may include, for example, source address information identifying the transmitter of the frame, destination address information identifying the intended recipient of some or all of the frame, routing information, identifier information identifying one or more aspects of some or all of the frame (e.g., information identifying the type of frame), control information, etc. The body may include, for example, a data payload (e.g., a payload of voice data, video data, web browsing data, application data, etc.). The trailer may include checking information that helps to verify the frame to the recipient. The checking information may include a frame check sequence (FCS) or cyclic redundancy check (CRC) field, as examples.

[0068] Under a bidirectional communications link between AP **104** and STA **106**, frames are conveyed both from AP **104** to STA **106** and from STA **106** to AP **104**. STA **106** may transmit one or more ACK frames or block ACK frames to AP **104** to acknowledge the successful receipt of one or more frames transmitted by AP **104**. AP **104** may transmit one or more ACK frames or block ACK frames to STA **106** to acknowledge the successful receipt of one or more frames transmitted by AP STA **106**.

[0069] When a first device (e.g., AP **104** or STA **106**) transmits wireless data to a second device (e.g., STA **106** or AP **104**) over a medium, care should be taken to ensure that the second device correctly receives the wireless data transmitted by the first device. In addition, care should be taken to prevent an unauthorized third device (e.g., a so-called man-in-the-middle (MITM) attacker) from receiving the wireless data transmitted by the first device, to prevent the third device relaying the wireless data or transmitting modified or other wireless data to the second device (e.g., while tricking the second device into believing that the third device is actually the first device), and/or to prevent the third device from tricking the second device into transmitting additional wireless data to the third device rather than the first device.

[0070] To help mitigate these issues, a first device may integrity protect one or more fields of the frames of wireless data transmitted by the first device to a second device (e.g., an intended recipient). The second device may receive the integrity-protected frames from the first device.

Upon receiving the integrity-protected frames, the second device may perform an integrity check (sometimes also referred to herein as an integrity verification or validation) to verify the integrity of the received integrity-protected frames. Verifying the integrity of the received integrity-protected frames may allow the second device to verify that it has correctly received all of the information in the integrity-protected frame transmitted by the first device and/or to verify that it has received the frame from the first device as opposed to an unauthorized third (e.g., MITM) device. The second device may also integrity protect one or more fields of the frames of wireless data that it transmits to the second device. The first device may then perform an integrity check on the integrity-protected frames transmitted by the second device.

[0071] FIG. 4 is a diagram showing how first and second devices in wireless communication system **108** may convey wireless data with each other. As shown in FIG. 4, wireless communication system **108** may include at least a first device **400A** (e.g., a STA **106** or AP **102/104** of FIG. 1) and a second device **400B** (e.g., a STA **106** or AP **102/104** of FIG. 1). Devices **400A** and **400B** may convey wireless data with each other over wireless communication link **404**.

Embodiments in which wireless communication link **404** is implemented and maintained using an IEEE 802.11 protocol (e.g., 802.11bn or other versions of the 802.11 protocol) are described herein as an example.

[0072] Device **400A** may integrity protect wireless data and may transmit the integrity-protected wireless data to device **400B** over wireless communication link **404**. The integrity-protected wireless data may include one or more frames (e.g., data frames). Device **400B** may perform an integrity check to verify data from the integrity-protected frame received from device **400A**. Device **400B** may transmit an acknowledgement message of the received integrity-protected frame, such as a block acknowledgement (BA) frame, to device **400A** over wireless communication link **404**. If desired, device **400B** may integrity protect the BA frame and device **400A** may perform an integrity check to verify the integrity-protected BA frame received from device **400B**. In this example, device **400A** operates as a transmitting (TX) device and device **400B** operates as a receiving (RX) device. This is illustrative and, at other times, device **400B** may operate as a TX device whereas device **400A** operates as an RX device. The integrity check may help to prevent a third device such as device **402** (e.g., a MITM device) from interfering with wireless communication link **404**.

[0073] The transmitting device may utilize a transmit (TX) buffer and a temporary buffer to transmit data to the receiving device, which receives and processes the data in a temporary buffer and a reorder buffer. For example, when device **400A** is acting as a transmitting device, device **400A** processes transmitted data using TX buffer **406** and temporary buffer **408**. When device **400B** is acting as a receiving device, device **400B** processes received data using temporary buffer **410** and reorder buffer **412**. Device **400B** may also have a TX buffer for use in transmitting data. Device **400A** may also have a reorder buffer for use in receiving data.

[0074] FIG. 5 is a timing diagram illustrating an exemplary frame **500** of integrity-protected wireless data transmitted by device **400A** to device **400B** and a corresponding integrity-protected BA frame **502** transmitted by device **400B** to device **400A** in response to frame **500**. As shown in FIG. 5, frame **500** (e.g., a data frame) includes several fields (e.g., as specified by the corresponding IEEE 802.11 protocol). The fields of frame **500** may include a preamble such as preamble **506** (e.g., one or more preamble fields), followed by a delimiter field such as aggregated media access control protocol data unit (A-MPDU) delimiter **508** (sometimes also referred to herein as MPDU delimiter **508**), followed by a header such as MAC header **510**, followed by a MAC header (HDR) message integrity check (MIC) field such as HDR MIC **512**, followed by a payload field such as media access control protocol data unit (MPDU) field **514**. MPDU field **514** may also include a MIC (not shown in FIG. 5 for the sake of clarity) for the payload data within MPDU field **514**.

[0075] FIG. 5 illustrates a simplest case in which frame **500** includes only a single MPDU for the

sake of clarity. If desired, frame **500** may be a physical protocol data unit (PPDU) frame that includes multiple MPDUs. Each MPDU may be preceded by a corresponding MPDU delimiter (e.g., A-MPDU delimiter **508**) in the PPDU frame. In these implementations, A-MPDU delimiter **508**, MAC header **510**, HDR MIC **512**, and MPDU field **514** may form a single A-MPDU subframe **530** (delimiter **508**, MAC header **510**, HDR MIC **512**, and MPDU field **514** are sometimes also referred to herein as A-MPDU subframe **530** or simply as subframe **530**). The PPDU frame may include a series of multiple A-MPDU subframes. The MPDUs in the frame may each have a respective sequence number (SN) identifying the temporal position or order of the MPDUs within the frame. The MAC header of each subframe may identify the SN of its corresponding MPDU along with other header information.

[0076] Preamble **506** may include a PHY layer preamble for frame **500** and/or other preamble information. A-MPDU delimiter **508** may define the position and length of its corresponding MPDU field **514** within frame **500** (e.g., inside an aggregated frame in implementations where frame **500** is a PPDU frame). MAC header **510** includes MAC header information for MPDU field **514**. As an example, MAC header **510** may include a header packet number (HDR PN) field. The HDR PN field may contain a unique value that is used to detect replay of the MAC header and in nonce of the integrity protection. HDR MIC **512** includes integrity protection check information for MAC header **510**. HDR MIC **512** may, for example, be generated by inputting some or all of MAC header **510** to a cryptographic function such as a hashing function (e.g., HDR MIC **512** may be a hash value or other cryptographic function output generated based on the content of MAC header **510**). MPDU field **514** may include any desired payload data for frame **500**. MPDU field **514** may also include an additional MIC (not shown) that is generated by inputting some or all of the data payload of MPDU field **514** into a cryptographic function such as a hashing function. While not shown in FIG. 5 for the sake of clarity, frame **500** may also include a final (trailing) field after its last MPDU that includes a frame check sequence (FCS). The FCS may be a checksum or another value calculated based on all of the data in frame **500** (e.g., the FCS may be an output from a hashing function, checksum function, or other cryptographic function that takes as its input all of the data of frame **500**).

[0077] Device **400B** (FIG. 4) may generate and transmit BA frame **502** in response to receipt of frame **500**. Device **400B** may transmit BA frame **502** after a time period such as short interframe space (SIFS) **504** has elapsed since the end of frame **500** or MPDU field **514** (e.g., as specified by the IEEE 802.11 protocol). SIFS **504** may, for example, provide time for device **400B** to process frame **500** and to respond with an ACK frame such as BA frame **502**. Device **400B** may transmit BA frame **502** using a predetermined or preconfigured duration, modulation coding scheme (MCS), and number of spatial streams (NSS).

[0078] As shown in FIG. 5, BA frame **502** includes several fields (e.g., as specified by the corresponding IEEE 802.11 protocol). BA frame **502** may include a preamble such as preamble **516**, a block acknowledgement (ACK) message field such as block ACK **518**, and a message integrity check field such as MIC **520**. Preamble **516** may include a PHY layer preamble for BA frame **502** and/or other preamble information. Block ACK **518** may include information acknowledging, for device **400A**, that device **400B** has successfully received the corresponding MPDU field(s) **514** in frame **500**. In implementations where frame **500** includes a single MPDU field **514**, BA frame **502** may sometimes also be referred to simply as an ACK frame. In implementations where frame **500** includes multiple MPDU fields **514** (e.g., when frame **500** is a PPDU frame), block ACK **518** may acknowledge the successful receipt of multiple MPDUs from frame **500** at device **400B**.

[0079] MIC **520** includes integrity protection check information for block ACK **518**. MIC **520** may, for example, be generated by inputting some or all of block ACK **518** to a cryptographic function such as a hashing function (e.g., MIC **520** may be a hash value or other cryptographic function output generated based on the content of block ACK **518**). Device **400A** is sometimes also referred

to herein as the transmitting or transmitter (TX) device of frame **500** and/or the recipient or receiver (RX) device of BA frame **502**. Device **400B** is sometimes also referred to herein as the TX device of BA frame **502** and/or the RX device of frame **500**.

[0080] FIG. **6** is a flow chart of illustrative operations involved in conveying frame **500** and BA frame **502** of FIG. **5** between devices **400A** and **400B** of FIG. **4**. At operation **600** of FIG. **6**, device **400A** may generate an integrity-protected frame such as frame **500**. Device **400A** may integrity-protect the frame by generating a respective HDR MIC **512** based on the MAC header **510** of each MPDU field **514** in frame **500**, for example. Device **400A** may then include HDR MICs **512** in frame **500**. Device **400A** may also integrity-protect the frame by generating an additional MIC for the data payload of each MPDU field **514** and inserting the additional MIC into or adjacent to each MPDU field. If desired, device **400A** may combine a HDR MIC with the corresponding additional MIC for a given MPDU field to produce a combined MIC that is inserted into the frame. Device **400A** may also generate an FCS based on the content of frame **500** and may include the FCS in frame **500** (e.g., in a trailing field of frame **500**). If desired, device **400A** may insert one or more padding fields into the frame to help accommodate processing delays.

[0081] At operation **602**, device **400A** may transmit frame **500** to device **400B**.

[0082] At operation **604**, device **400B** may begin receiving frame **500** from device **400A**. Device **400B** may integrity check (e.g., may verify the integrity of) the received frame **500**. This may sometimes also be referred to herein as validating the integrity of frame **500**, validating the integrity protection of frame **500**, verifying the integrity of frame **500**, verifying the integrity protection of frame **500**, checking the integrity of frame **500**, or checking the integrity protection of frame **500**. If desired, the integrity check may contain a replay detection operation. Replay detection may be performed to ensure that the HDR PN values are increasing over time (e.g., by comparing a HDR PN value to a previous HDR PN value and determining whether the newer HDR PN value is greater than the previous HDR PN value). The integrity check may also include computing, calculating, or generating a HDR MIC for each MAC header **510** in the received frame (e.g., using the same hashing function or cryptographic function used by device **400A** to generate HDR MICs **512** with the corresponding MAC header **510** as an input to the function) and comparing the calculated HDR MIC to the HDR MICs **512** included in frame **500**. If desired, this may also include computing, calculating, or generating an additional MIC for the data payload in each MPDU field **514** of the received frame and comparing the calculated additional MIC(s) to the corresponding additional MIC(s) of the MPDU field(s) **514** as included in frame **500**. If desired, device **400B** may also generate an FCS for the received frame **500** and may compare the generated FCS to an FCS included in frame **500**. Device **400B** successfully verifies/validates the integrity of the received frame **500** (e.g., the integrity check is successful) if/when each of the calculated HDR MICs matches the corresponding HDR MICs **512** included in frame **500** and/or each of the calculated additional MICs matches the corresponding additional MICs in frame **500**. If one of the calculated HDR MICs does not match the corresponding HDR MIC **512** included in frame **500** and/or one of the calculated additional MICs does not match the corresponding additional MIC included in frame **500**, device **400B** does not verify the integrity of received frame **500** (e.g., the integrity check is unsuccessful/failed, the frame is identified as invalid or unverified, etc.). Device **400B** may take appropriate action when the integrity check is successful or unsuccessful.

[0083] At operation **606**, device **400B** may generate an integrity-protected BA frame such as BA frame **502**. Device **400B** may integrity-protect the BA frame by generating MIC **520** based on block ACK **518** and including MIC **520** in BA frame **502**. If desired, device **400B** may insert one or more padding fields into the BA frame to help accommodate processing delays.

[0084] At operation **608**, device **400B** may transmit the integrity-protected BA frame to device **400A**. Device **400B** may perform operations **606** and/or **608** prior to or concurrently with operation **604** if desired.

[0085] At operation **610**, device **400A** may begin receiving BA frame **502** from device **400B**.

Device **400A** may integrity check (e.g., may verify the integrity of) the received BA frame **502**. This may sometimes also be referred to herein as validating the integrity of BA frame **502**, validating the integrity protection of BA frame **502**, verifying the integrity of BA frame **502**, verifying the integrity protection of BA frame **502**, checking the integrity of BA frame **502**, or checking the integrity protection of BA frame **502**. This may include computing, calculating, or generating a MIC for block ACK **518** in the received BA frame (e.g., using the same hashing function or cryptographic function used by device **400B** to generate MIC **520** with the corresponding block ACK **518** as an input to the function) and comparing the generated MIC to the MIC **520** included in BA frame **502**. The integrity check may also optionally include soliciting frame verification, described in greater detail below. Device **400A** successfully verifies/validates the integrity of the received BA frame **502** (e.g., the integrity check is successful) if/when the generated MIC matches the MIC **520** included in BA frame **502**. If the generated MIC does not match the MIC **520** included in BA frame **502**, device **400A** does not verify the integrity of received frame **500** (e.g., the integrity check is unsuccessful). Device **400A** may take appropriate action when the integrity check is successful or unsuccessful. Processing may then loop back to operation **600** via path **612** as device **400A** continues to transmit wireless data to device **400B**. Device **400A** may perform operation **600** after or concurrently with a subsequent iteration of operation **600** if desired.

[0086] Integrity checking frame **500** and BA frame **502** (FIG. 5) in this way may help to protect against MITM attacks by other devices such as device **402** of FIG. 4 and may help to optimize wireless communication between devices **400A** and **400B**. MAC header integrity protection (e.g., HDR MICs **512** in frame **500**) may, for example, help to ensure that an attacker (e.g., device **402**) cannot change a STA **106** (FIG. 1) to receive PPDUs with excessive bandwidth (BW) or NSS (e.g., via operating mode indication (OMI) A-Control) and/or may help to ensure that an attacker cannot cause AP **102/104** (FIG. 1) to believe that STA **106** is incorrectly in a power save or active mode, which could otherwise cause frame loss or a lack of frame transmission (e.g., via the STA's Power Management field). On the other hand, BA frame integrity protection (e.g., MIC **520** in BA frame **502**) may help to ensure that the SN of devices **400A** and **400B** are synchronized. For example, if device **400B** is expecting an excessively high SN, device **400B** can drop received frames with smaller SNs. In addition, if device **400B** has a hole in its reorder buffer, device **400B** may not properly forward received frames to its corresponding software application or to the network **100** (FIG. 1), which can cause transmission delays.

[0087] Consider one example of a MITM attack performed by device **402** of FIG. 4. In this example, device **402** intercepts a frame **500** transmitted by device **400A** (e.g., a frame having MPDUs with sequence numbers (SN) **1**, **2**, **3**, and **4**), device **402** modifies the data from frame **500** before transmitting the modified data to device **400B** (e.g., a frame having only MPDUs with sequence numbers **1** and **2** but not sequence numbers **3** and **4** from the data transmitted by device **400A**), and device **400A** does not receive a BA frame. Integrity protection and checking can help to mitigate this type of attack.

[0088] Integrity checking frame **500** and BA frame **502** may also require processing delays for device **400A** and **400B**. For example, returning to FIG. 5, device **400A** may require a time period **522** to generate (e.g., calculate, compute, etc.) HDR MIC **512** for MAC header **510** (e.g., at operation **600** of FIG. 6). As another example, device **400B** may require a time period **526** following receipt of the end of MAC header **510** to verify HDR MIC **512** (e.g., at operation **604** of FIG. 6). As yet another example, device **400B** may require a time period **524** to generate (e.g., calculate, compute, etc.) MIC **520** for block ACK **518** (e.g., at operation **606** of FIG. 6). As still another example, device **400A** may require a time period **528** following receipt of the end of MIC **520** to verify MIC **520** (e.g., at operation **610** of FIG. 6).

[0089] If desired, frame **500** and/or BA frame **502** may include one or more padding fields that help to help accommodate these processing delays. FIGS. 7A-D show four examples of padding

fields that device **400A** may include in frame **500**. As shown in the first example of FIG. 7A, device **400A** may generate a frame **500** that includes a padding field such as padding (PAD) **700** between A-MPU delimiter **508** and MAC header **510**. In general, padding **700** may include any desired sequence of one or more padding bits. As one example, padding **700** may include one or more repetitions of A-MPDU delimiter **508** (e.g., there may be two or more copies of A-MPDU delimiter **508** between preamble **506** and MAC header **510**).

[0090] In another example, as shown in FIG. 7B, padding **700** may be between MAC header **510** and HDR MIC **512** in frame **500**. In yet another example, as shown in FIG. 7C, padding **700** may be inserted between HDR MIC **512** and MPDU field **514** in frame **500**. In a further example, as shown in FIG. 7D, padding **700** may be appended to the end of MPDU field **514** and/or frame **500**. Padding **700** of FIGS. 7A-7C may include any desired sequence of one or more padding bits or symbols (e.g., orthogonal frequency division multiplexing (OFDM) symbols that each have a 4 us duration, a 12 us duration, or another duration as specified by the communications protocol). If desired, padding **700** may be appended to the beginning of preamble **506** in some cases (e.g., at location **702**). If desired, frame **500** may include padding **700** at two or more of the locations shown in FIGS. 7A-7D. As examples, padding **700** may include one or more repetitions of A-MPDU delimiter **508**, repetitions of one or more other fields of frame **500**, empty or null fields, a field or series of bits or symbols defined by the communications protocol as being intended or reserved for padding, etc. Padding **700** may, for example, allow more time for device **400A** to generate HDR MIC **512** for MAC header **510** when generating and transmitting frame **500** and/or for device **400B** to verify HDR MIC **512** in the received frame **500**.

[0091] FIG. 8 is a diagram showing one example in which frame **500** is a PPDU frame having multiple MPDUs. As shown in FIG. 8, frame **500** may include a set of MPDU fields **800**. Each MPDU field **800** may include a respective MPDU field **514** of FIG. 5 and may, if desired, be associated with a different respective subframe (e.g., subframe **530** of FIG. 5). If desired, each MPDU field **800** may also include one or more of the other fields **506**, **508**, **510**, and **512** of FIG. 5 before the corresponding MPDU field **514**.

[0092] Each MPDU field **800** has a corresponding SN specifying the order of the MPDU fields within frame **500**. In the example of FIG. 8, frame **500** has four MPDU fields **800**, including a first MPDU field **800-0** with SN=A, a second MPDU field **800-1** with SN=A+1, a third MPDU field **800-2** with SN=A+2, and a fourth MPDU **800-3** with SN=A+3. MPDU fields **800-0** through **800-3** may also have respective increasing HDR PN values. This is illustrative and, in general, frame **500** may have any desired number of MPDU fields **800**.

[0093] Frame **500** may include padding **700** before and/or after each MPDU field **800**. If desired, frame **500** may have different paddings **700** with different characteristics and/or including different padding bits and/or symbols at different locations within the frame. For example, there may be a first set of padding fields **700A** and a second set of padding fields **700B** in frame **500**. Padding fields **700A** may, for example, occur before and after each MPDU field **800** in frame **500** except for the final MPDU field **800** in frame **500** (e.g., MPDU field **800-3**). Padding fields **700B** may occur after the final MPDU field **800** in frame **500** (e.g., MPDU field **800-3**). In the example of FIG. 8, there is a single padding field **700A** before MPDU field **800-0** and a set of ten padding fields **700A** between MPDU fields **800-0** and **800-1** and between MPDU fields **800-1** and **800-2**, six padding fields **700A** between MPDU fields **800-2** and **800-3**, and six padding fields **700B** after MPDU field **800-3**. This is illustrative and, in general, frame **500** may include any desired number of padding fields **700A** and **700B**.

[0094] In some implementations, each padding field **700A** may include a duplicated A-MPDU delimiter **508** (FIG. 5) (e.g., padding fields **700A** may be duplicates of the same MPDU delimiter). A duplicated A-MPDU delimiter may be, for example, a zero-length A-MPDU subframe with its end of frame (EOF) bit set to 0. In some implementations, each padding field **700B** may include EOF padding bits (e.g., a zero length A-MPDU subframe with its EOF bit set to 1). There may be

sufficient padding fields **700B** (e.g., EOF bits in frame **500**) to meet a minimum EOF padding duration **804** for frame **500**. Padding fields **700A** may help to ensure that a minimum MPDU start spacing **802** exists between the beginning of each MPDU field **800** in frame **500** and the beginning of the subsequent MPDU field **800** in frame **500**. Device **400A** may adjust or re-configure the number of padding fields **700A** and/or **700B** to allow more or less time for device **400A** to generate frame **500** and/or for device **400B** to integrity check frame **500**.

[0095] Minimum MPDU start spacing **802** represents the time in which one MPDU field **800** may be transmitted by device **400A**, regardless of the size of the MPDU field. Additional padding fields **700A** (e.g., A-MPDU subframe headers, MPDU delimiters, etc.) may be added to ensure that the next MPDU field **800** begins only after minimum MPDU start spacing **802**. If MPDU transmission takes longer than minimum MPDU start spacing **802** then device **400A** may forego the inclusion of padding fields, as padding is not needed. The added time from padding fields **700A** and **700B** may be in units of symbols, for example.

[0096] Minimum EOF padding duration **804** is the amount of time when padding is transmitted at the end of or after frame **500**. The EOF padding duration (e.g., the number of padding fields **700B**) may increase the number of symbols that need to be transmitted at the end of or after the data frame. The receiver (e.g., device **400B**) may use this time to integrity check the data frame. The receiver may also use this time to create the integrity-protected BA frame **502** (FIG. 5).

[0097] FIGS. **9A** and **9B** show two examples of padding fields that device **400B** may include in BA frame **502** of FIG. 5. As shown in the example of FIG. **9A**, device **400B** may generate a BA frame **502** that includes a padding field such as padding **900** between preamble **516** and block ACK **518**. In this example, padding **900** may include similar padding as padding fields **700A** of FIG. **8** (e.g., one or more duplicated MPDU delimiters) or may include other padding. In another example, as shown in FIG. **9B**, padding **900** may be inserted between block ACK **518** and MIC **520**. If desired, similar padding to padding **900** may also be included within block ACK **518** (e.g., as additional Per Association Identifier (AID) fields). If desired, BA frame **502** may include padding **900** at two or more of these locations. In general, padding **900** may include similar padding as padding fields **700A** of FIG. **8** (e.g., a duplicated MPDU delimiter), one or more AID traffic identifier (TID) Info fields that are not used by any STA **106**, or other padding bits or symbols. If desired, the receiver may help to configure the minimum EOF padding duration **804** used to transmit frame **500** (FIG. **8**) to allow the receiver sufficient time to calculate and generate the BA frame.

[0098] Device **400B** may transmit BA frame **502** to a receiver of the BA frame (e.g., device **400A**). BA frame **502** may have a minimum duration **902**. Minimum duration **902** plus the minimum EOF padding duration **804** (FIG. **8**) may signal to an AP the minimum duration of the BA frame that the receiver should consider to receive. If the BA frame includes relatively large Per AID fields, then this duration may be exceeded. If the AP triggers uplink (UL) BA frames by sending a corresponding trigger frame, then the UL reservation unit duration for the BA frame may be at least the minimum duration **902** plus minimum EOF padding duration **804** (FIG. **8**). Minimum duration **902** may be replaced with the minimum EOF padding time, for example. In this case, all extra time is allocated to frame **500**. If desired, extra padding time may be added after the end of MIC **520** in BA frame **502** (e.g., as shown by padding **900** of FIG. **9B**). This time is used by the transmitter (e.g., device **400A**) to verify the integrity of BA frame **502** and to check whether to continue data transmissions.

[0099] Padding **700** may be added to frame **500** and/or padding **900** may be added to BA frame **502** (e.g., before block ACK **518**, after block ACK **518**, and/or within block ACK **518** as additional Per AID fields) to accommodate integrity protection and integrity check processing delays. When padding **700** is added to the end of frame **500** (see, e.g., FIGS. **7D** and **8**), the padding duration is configured in the association and the padding is present in all transmitted A-MPDUs. This may require additional setup signaling. As the padding is always present, this type of padding always involves additional overhead. On the other hand, when padding **900** is added to BA frame **502**, the

BA frame transmitter (e.g., device **400B**) may add the padding as per-Association Identifier (AID) padding fields if/when the calculation of MIC **520** requires additional time. Padding **900** may be, for example, one or more AID TID Info fields that are not used by any STA **106**. This implementation does not require any signaling overhead to set up and the BA frame transmitted may simply apply the padding when needed.

[0100] Consider an example in which device **400A** performs transmit opportunity (TXOP) continuation after receipt of a BA frame **502**. In these situations, device **400A** does not necessarily know whether the true (intended) receiver (e.g., device **400B**) has received its transmitted frame **500** until device **400A** has verified the integrity of the corresponding BA frame **502** transmitted by device **400B**. In this situation, device **400A** may verify the integrity of BA frame **502** (e.g., having padding **900** at the trailing end of the BA frame as shown in FIG. 9B) before or after continuing TXOP by transmitting additional frames **500** to device **400B**.

[0101] In implementations where device **400A** begins to transmit additional frames **500** regardless of whether device **400A** has validated the integrity of BA frame **502**, device **400A** estimates the duration of the next PPDU without ensuring that frames are received by the intended receiver. If an attacker (e.g., device **402** of FIG. 4) transmitted the BA frame, the next PPDU duration may be allocated incorrectly, leading to communication errors. By adding padding **900** to BA frame **502**, device **400B** may allow more time for device **400A** to verify (validate) the integrity of BA frame **502**. By verifying the integrity of BA frame **502**, device **400A** knows that its transmitted frame **500** was received by the true (intended) receiver (device **400B**) as well as the number of MPDUs received by the true receiver. On the other hand, padding **900** adds transmission overhead for device **400B**.

[0102] The header(s) in frame **500** may include a corresponding Header (HDR) Packet Number (PN) (HDR PN), a time synchronization function (TSF) component (e.g., specifying the time when the corresponding preamble is to begin transmission), and HDR MIC (e.g., HDR MIC **512** of FIG. 5). HDR PN may include a unique counter value that is incremented by one for each integrity-protected header transmitted by device **400A**. Device **400B** may use several protection mechanisms to prevent the reception of fake data from a MITM (e.g., device **402**) and the transmission of a BA frame as a response to the receipt of fake data. For example, device **400B** may perform an address and FCS check. Device **400B** may transmit BA frame **502** if frame **500** is properly addressed to device **400B** and the FCS computed at device **400B** in response to the received frame **500** matches the FCS field of the received frame **500**. Additionally or alternatively, device **400B** may check the header PN(s) and may transmit BA frame **502** if the MPDU is not replayed. Additionally or alternatively, device **400B** may check the header TSF(s) and HDR PNs and may transmit BA frame **502** if the TSF is within a predetermined time period (e.g., approximately 256 us) and the HDR PN increasing counter value together with the TSF (if present) have a value that is larger than a previously received value for a correctly integrity protected frame. Additionally or alternatively, device **400B** may calculate the HDR MIC(s) and may transmit BA frame **502** if the calculated HDR MIC(s) match the header MIC(s) included in frame **500**. Additionally or alternatively, device **400B** may calculate the MIC for data payload(s) in frame **500** and may transmit BA frame **502** if the calculated MIC(s) match the MIC(s) for the data payload(s) included in frame **500**. The integrity check at device **400B** may occur before or after device **400B** transmits BA frame **502**, with differing implications for how device **400A** and **400B** process the transmitted and received data.

[0103] In general, when device **400B** transmits BA frame **502** regardless of MAC header integrity validation for frame **500**, the A-MPDU does not need extra padding. Device **400B** may transmit the BA frame when the MAC headers A1 and A2 match and device **400B** has correctly received at least one MPDU (e.g., the FCS matches). On the other hand, when device **400B** transmits BA frame **502** only if/when MAC header integrity is validated, extra padding may be needed to accommodate the MAC header integrity check (sometimes also referred to herein as MAC header integrity validation). Device **400B** may transmit the BA frame when MAC headers A1 and A2 match and

device **400B** receives at least one MPDU, successfully verifies the integrity protection of its MAC header, and successfully verifies that it has correctly received the MPDU (e.g., the FCS matches). [0104] Buffers on device **400A** and device **400B** (e.g., buffers **406-412** of FIG. **4**) may be adapted to handle the transmission, reception, and validation of frame **500** and BA frame **502**. For example, when device **400A** has data to transmit to device **400B**, device **400A** may select one or more MPDUs for transmission to device **400B**. Device **400A** may aggregate the MPDUs into frame **500** (e.g., at TX buffer **406** and/or temporary buffer **408** of FIG. **4**). Device **400A** may integrity protect the MPDUs and/or frame. For example, device **400A** may integrity protect the MAC headers **510** of the MPDU fields **514** in frame **500** by generating corresponding HDR MICs **512** and inserting HDR MICs **512** into frame **500**. Device **400A** may also encrypt the data payloads of the MPDUs and/or may generate and insert additional MICs of the data payloads of the MPDUs.

[0105] Device **400B** may receive frame **500** and may store frame **500** in temporary buffer **410**. Device **400B** may perform an integrity check on the MPDUs stored in temporary buffer **410**. Device **400B** may, for example, only acknowledge a received MPDU (e.g., in a corresponding BA frame **502**) when the MPDU has all of the following: (a) matching MAC headers A1 and A2, (b) MAC header numbers (MHNs) that are in increasing order, and (c) payloads with correct FCSs. Device **400B** may update reorder buffer **412** (FIG. **4**) to only include MPDUs or frames **500** from temporary buffer **410** that are valid (e.g., that pass the integrity check). For example, Device **400B** may validate the integrity of MPDUs in temporary buffer **410** by performing one or more of the following: (i) validating the integrity of the MAC header for each MPDU in temporary buffer **410** (e.g., by calculating a HDR MIC from the MAC header and comparing it to the corresponding HDR MIC **512** received in frame **500** and stored in the temporary buffer), (ii) decrypting the data payload for each MPDU and integrity checking the decrypted data payload, and (iii) checking whether reorder buffer **412** already contains the PN of the corresponding frame **500**. Device **400A** may update its TX buffer **406** (FIG. **4**) based on integrity checked BA frames **502** received from device **400B**. Device **400A** may, for example, receive a BA frame **502** from device **400B** and may store the BA frame in temporary buffer **408** (FIG. **4**). Device **400A** may then verify the integrity of the BA frame in temporary buffer **408** (e.g., may check whether the bitmap of the BA frame matches the currently transmitted MPDUs in temporary buffer **408**). If both conditions are met, device **400A** may update its TX buffer **406** based on the now-verified (integrity-checked) BA frame **502**.

[0106] FIGS. **10-12** are flow charts of illustrative operations that may be performed by device **400B** to integrity check a frame **500** transmitted by device **400A** before or after device **400B** transmits the corresponding BA frame **502** to device **400A**. The operations of FIGS. **10-12** may, for example, be performed while processing operation **604** and **606** of FIG. **6**.

[0107] FIG. **10** is a flow chart of operations that may be performed by device **400B** in implementations where device **400B** integrity checks the MAC header MIC (e.g., HDR MIC **512** of FIG. **5**) and the corresponding FCS of frame **500** after generating and transmitting BA frame **502**.

[0108] At operation **1000**, device **400B** may detect (e.g., may receive and begin attempting to decode/demodulate) a frame **500** transmitted by device **400A**. Frame **500** includes at least one MPDU field **514** (FIG. **5**). The remaining operations of FIG. **10** may be performed on a given MPDU field **514** from frame **500**. In situations where frame **500** includes multiple MPDU fields **514** (e.g., for MPDU fields **800** in a single PPDU as shown in FIG. **8**), the operations of FIG. **10** may be repeated for each of the MPDUs in the received frame. Device **400B** may temporarily store the MPDU at temporary buffer **410** (FIG. **4**) for further processing.

[0109] At operation **1002**, device **400B** may check, verify, validate, identify, and/or otherwise determine whether device **400B** is the intended recipient of the MPDU field **514** from frame **500**. This may include, for example, determining whether the AID and destination address of the MPDU stored on the temporary buffer (e.g., as identified by one or more header fields of the MPDU) are associated with device **400B** or another recipient device. If/when device **400B** is not the intended

recipient of the MPDU (e.g., when the AID and/or address do not match), processing may proceed to operation **1006** via path **1004**.

[0110] At operation **1006**, device **400B** may discard (delete) the MPDU from the temporary buffer and processing may loop back from operation **1006** to operation **1002** to process additional MPDUs of the detected frame **500**. Alternatively, processing may loop back from operation **1006** to operation **1000** to process additional frames **500** received from device **400A**. If/when device **400B** is the intended recipient of the MPDU (e.g., when the AID and address match), processing may proceed from operation **1002** to operation **1010** via path **1008**.

[0111] At operation **1010** (e.g., responsive to device **400B** being the intended recipient of the MPDU), device **400B** may check, verify, and/or validate a MAC header number (MHN) and TSF of the MPDU stored at the temporary buffer. The MHN and the TSF may, for example, be identified by or included within the MAC header **510** (FIG. 5) of the MPDU. Device **400B** may check the MHN and TSF by, for example, determining whether the MHN is a duplicate MHN and whether the TSF falls within a predetermined range. Device **400B** may determine or identify that the MHN is valid if/when the MHN is not a duplicate MHN and may determine that the TSF is valid if/when the TSF is within a predetermined time range. On the other hand, device **400B** may determine or identify that the MHN is invalid if/when the MHN is a duplicate MHN and may determine that the TSF is invalid if/when the TSF is outside the predetermined time range. If/when one or both of the MHN and TSF are invalid, processing proceeds to operation **1006** via path **1020** and the MPDU may be discarded. If/when the MHN and the TSF are valid, processing proceeds from operation **1010** to operation **1016** via path **1014**.

[0112] At operation **1016** (e.g., responsive to device the MPDU having a valid MHN and TSF), device **400B** check, verify, and/or validate a payload FCS of the MPDU stored at the temporary buffer. This may include calculating an FCS based on the payload of the MPDU and comparing the calculated FCS to an FCS field of the MPDU. Device **400B** may determine or identify that the FCS is valid when the calculated FCS matches the FCS identified by the FCS field of the MPDU. Device **400B** may determine or identify that the FCS is invalid when the calculated FCS does not match the FCS identified by the FCS field of the MPDU. If/when the FCS is invalid, processing proceeds to operation **1006** via path **1018** and the MPDU may be discarded. If/when the FCS is valid, processing proceeds from operation **1016** to operation **1022** via path **1020**.

[0113] At operation **1022** (e.g., responsive to the MPDU having a valid FCS), device **400B** may prepare (generate) a BA frame **502** having a block ACK **518** (FIG. 5) that includes and/or otherwise identifies the corresponding MPDU received in frame **500** (e.g., the MPDU as checked at operations **1008-1016**). Device **400B** may transmit BA frame **502** to device **400A**.

[0114] At operation **1024** (e.g., after transmission of BA frame **502**), device **400B** may decrypt the MAC header **510** (FIG. 5) and the data payload of the MPDU from the detected frame **500**. This may include checking, verifying, and/or validating the HDR MIC **512** of the MPDU as well as checking, verifying, and/or validating a MIC field of the data payload of the MPDU.

[0115] Device **400B** may validate HDR MIC **512** by calculating a header MIC based on the MAC header **510** of the MPDU (e.g., by applying the same hashing function or cryptographic function to MAC header **510** as applied by device **400A** when generating HDR MIC **512**) and by comparing the calculated header MIC to the HDR MIC **512** of the MPDU as included in the received frame **500** (e.g., device **400B** may determine that HDR MIC **512** is valid when the calculated header MIC matches the HDR MIC **512** included with the MPDU and may determine that HDR MIC **512** is invalid when the calculated header MIC does not match the HDR MIC **512** included with the MPDU).

[0116] Device **400B** may validate the MIC field of the data payload of the MPDU by calculating a MIC based on the data payload of the MPDU (e.g., by applying the same hashing function or cryptographic function to the data payload as applied by device **400A** when generating the MPDU) and comparing the calculated MIC to the MIC for the data payload as included with the MPDU in

the detected frame **500** (e.g., device **400B** may determine that MIC for the data payload is valid when the calculated MIC matches the MIC for the data payload as included with the MPDU and may determine that the MIC for the data payload is invalid when the calculated MIC does not match the MIC for the data payload as included with the MPDU).

[0117] If/when device **400B** determines that HDR MIC **512** is invalid, that the MIC for the data payload is invalid, or that device **400B** is otherwise unable to recover or decrypt the data payload, processing may proceed to operation **1006** via path **1026**. If/when device **400B** determines that HDR MIC **512** is valid, that the MIC for the data payload is valid, and device **400B** is able to recover or decrypt the data payload, processing proceeds to operation **1030** via path **1028**.

[0118] At operation **1030**, device **400B** may reorder buffer the MPDU from the detected frame **500**. This may include passing the MPDU from temporary buffer **410** to reorder buffer **412** (FIG. 4). Recorder buffer **412** may reorder stored MPDUs (e.g., in the correct time order as identified by the SN of the MPDUs transferred onto reorder buffer **412** from temporary buffer **410**).

[0119] At operation **1032**, the reorder buffer **412** on device **400B** (FIG. 4) may deliver the MPDU(s) stored on reorder buffer **412** to one or more software applications running on device **400B** for further processing (e.g., device **400B** may pass the stored MPDU(s) up the protocol stack to an applications processor on device **400B**).

[0120] The example of FIG. 10 in which device **400B** transmits BA frame **502** before checking the MIC of the data payload and the HDR MIC **512** for the MAC header **510** in the MPDU stored at temporary buffer **410** is illustrative and non-limiting. If desired, device **400B** may check the HDR MIC **512** for the MAC header **510** of the MPDU prior to transmitting BA frame **502**.

[0121] FIG. 11 is a flow chart of illustrative operations that may be performed by device **400B** in implementations where device **400B** integrity checks MAC header **510** of the MPDU prior to transmitting BA frame **502**. Operations **1100-1116** of FIG. 11 are the same as operations **1000-1016** of FIG. 10, respectively. Description of operations **1100-1116** of FIG. 11 have therefore been omitted for the sake of brevity.

[0122] At operation **1122** (e.g., prior to transmitting BA frame **502** and responsive device **400B** determining that the FCS payload is valid), device **400B** may validate HDR MIC **512** by calculating a header MIC based on the MAC header **510** of the MPDU stored on the temporary buffer and by comparing the calculated header MIC to the HDR MIC **512** included with the MPDU in the detected frame **500**. Device **400B** may determine that HDR MIC **512** is valid when the calculated header MIC matches the HDR MIC **512** included with the MPDU and may determine that HDR MIC **512** is invalid when the calculated header MIC does not match the HDR MIC **512** included with the MPDU.

[0123] If/when device **400B** determines that HDR MIC **512** is invalid, processing may proceed to operation **1106** via path **1119**. If/when device **400B** determines that HDR MIC **512** is valid, processing proceeds to operation **1124** via path **1123**. At operation **1124** (e.g., responsive to the MPDU having a valid HDR MIC **512**), device **400B** may prepare (generate) a BA frame **502** having a block ACK **518** (FIG. 5) that includes and/or otherwise identifies the corresponding MPDU received in frame **500** (e.g., the MPDU as checked at operations **1102-1122**). Device **400B** may transmit BA frame **502** to device **400A**.

[0124] At operation **1126** (e.g., after transmission of the BA frame), device **400B** may validate the MIC field of the data payload of the MPDU by calculating a MIC based on the data payload of the MPDU and comparing the calculated MIC to the MIC for the data payload as included with the MPDU in the detected frame **500**. Device **400B** may determine that MIC for the data payload is valid when the calculated MIC matches the MIC for the data payload as included with the MPDU and may determine that the MIC for the data payload is invalid when the calculated MIC does not match the MIC for the data payload as included with the MPDU.

[0125] If/when device **400B** determines that the MIC for the data payload is invalid, processing may proceed to operation **1106** via path **1128**. If/when device **400B** determines that the MIC for the

data payload is valid, and device **400B** may recover or decrypt the data payload and processing proceeds to operation **1130** via path **1129**. Operations **1130** and **1132** of FIG. **11** are the same as operations **1030** and **1032** of FIG. **10**, respectively. Description of operations **1130** and **1132** have therefore been omitted for the sake of brevity.

[0126] If desired, device **400B** may check both the HDR MIC **512** for the MAC header **510** of the MPDU and the MIC for the data payload of the MPDU prior to transmitting BA frame **502**. FIG. **12** is a flow chart of illustrative operations that may be performed by device **400B** in implementations where device **400B** integrity checks both MAC header **510** and the data payload of the MPDU prior to transmitting BA frame **502**. Operations **1200-1216** of FIG. **12** are the same as operations **1000-1016** of FIG. **10**, respectively. Description of operations **1200-1216** of FIG. **12** have therefore been omitted for the sake of brevity.

[0127] At operation **1222** (e.g., prior to transmitting BA frame **502** and responsive device **400B** determining that the FCS payload is valid), device **400B** may validate HDR MIC **512** by calculating a header MIC based on the MAC header **510** of the MPDU stored on the temporary buffer and by comparing the calculated header MIC to the HDR MIC **512** included with the MPDU in the detected frame **500**. Device **400B** may determine that HDR MIC **512** is valid when the calculated header MIC matches the HDR MIC **512** included with the MPDU and may determine that HDR MIC **512** is invalid when the calculated header MIC does not match the HDR MIC **512** included with the MPDU.

[0128] If/when device **400B** determines that HDR MIC **512** is invalid, processing may proceed to operation **1206** via path **1219**. If/when device **400B** determines that HDR MIC **512** is valid, processing proceeds to operation **1224** via path **1223**. At operation **1224** (e.g., responsive to the MPDU having a valid HDR MIC **512**), device **400B** may validate the MIC field of the data payload of the MPDU by calculating a MIC based on the data payload of the MPDU and by comparing the calculated MIC to the MIC for the data payload as included with the MPDU in the detected frame **500**. Device **400B** may determine that MIC for the data payload is valid when the calculated MIC matches the MIC for the data payload as included with the MPDU and may determine that the MIC for the data payload is invalid when the calculated MIC does not match the MIC for the data payload as included with the MPDU. If desired, operation **1222** may be performed concurrently with operation **1224** (e.g., in a single combined integrity check step with operation **1224**) or prior to operation **1224**.

[0129] If/when device **400B** determines that the MIC for the data payload is invalid, processing may proceed to operation **1206** via path **1226**. If/when device **400B** determines that the MIC for the data payload is valid, and device **400B** may recover or decrypt the data payload and processing proceeds to operation **1230** via path **1228**. At operation **1230** (e.g., responsive to the MPDU having both a valid HDR MIC **512** and a valid data payload MIC), device **400B** may prepare (generate) a BA frame **502** having a block ACK **518** (FIG. **5**) that includes and/or otherwise identifies the corresponding MPDU received in frame **500** (e.g., the MPDU as checked at operations **1202-1224**). Device **400B** may transmit BA frame **502** to device **400A**. Operations **1232** and **1234** of FIG. **12** are the same as operations **1030** and **1032** of FIG. **10**, respectively. Description of operations **1232** and **1234** have therefore been omitted for the sake of brevity.

[0130] Device **400A** generally has no knowledge when device **400B** discards a frame or MPDU after transmission of BA frame **502** (e.g., as shown by path **1128** of FIG. **11** or path **1026** of FIG. **10**). This may cause the receiver to miss such a frame. If desired, portions of the MAC header (e.g., a power management (PM) field, an end of service period (EOSP) field, a high throughput (HT) control field, and/or mobility domain (MD) bits) may be integrity protected and integrity checked during one or more of the checks of FIGS. **10-12** and/or in separate operations.

[0131] The examples of FIGS. **10-12** are illustrative and non-limiting. In general, device **400B** may implement other integrity check procedures. Under any of these schemes, when the FCS, header PN, TSF, MAC header, and data payload have been validated for a given MPDU, device **400B** may

pass the MAC header and the data payload of the MPDU to its reorder buffer and may transmit BA frame **502** to device **400A** with or without integrity protection and/or device **400A** integrity checking the BA frame. When the FCS, header PN, TSF, and MAC header are valid but the payload MIC is invalid, neither the payload nor the MAC header are passed to the reorder buffer but device **400B** may still transmit BA frame **502** to device **400A** with or without integrity protection and/or device **400A** integrity checking the BA frame. Alternatively, in these situations, device **400B** may pass the MAC header to the reorder buffer when the HDR MIC of the MAC header is valid (e.g., when the header MIC is validated before validating the payload MIC). When FCS, header PN, and TSF are valid but the MAC header is invalid, neither the payload nor the MAC header are passed to the reorder buffer whether or not the data payload MIC is valid. Device **400B** may still transmit BA frame **502** to device **400A** without an integrity check but not with an integrity check. When the FCS, header PN, or TSF are invalid, neither the payload nor the MAC header are passed to the reorder buffer regardless of the MAC header integrity or the data payload integrity and device **400B** does not transmit a BA frame **502** to device **400A**.

[0132] Consider an example in which device **400B** transmits an integrity-protected BA frame **502** to device **400A** in response to a frame **500** having at least a first MPDU with SN=1, a second MPDU with SN=2, and a power management field of zero (e.g., PowerManagement=0). Frame **500** may solicit a corresponding BA frame **502** (e.g., the BA frame **502** transmitted by device **400B** is a solicited BA frame). In some instances, BA frame **502** may be intercepted by device **402** (FIG. 4) (e.g., a MITM attacker) rather than being received at device **400A**. In these situations, after a predetermined time period has elapsed without receipt of a BA frame corresponding to frame **500** at device **400A**, device **400A** may re-transmit frame **500** but with a power management field of one (e.g., PowerManagement=1). Device **402** may then intercept the re-transmitted frame. In response to intercepting the re-transmitted frame, device **402** may transmit (replay), to device **400A**, the BA frame **502** received from device **400B**. The replayed BA frame represents an insecure MITM transmission and interference in communications between devices **400A** and **400B**. Device **400A** may integrity check the received BA frame **502** and may pass the received BA frame out of its temporary buffer when device **400A** confirms that the received BA frame **502** is valid (e.g., while processing operation **610** of FIG. 6).

[0133] In these situations, device **400A** may utilize a number of mechanisms to avoid being tricked into believing that a BA frame re-played by device **402** was actually transmitted by device **400B**. First, device **400A** may check that the addresses and FCS of the received BA frame match the BA frame as solicited by frame **500**. Second, device **400A** may check the PN of the received BA frame to ensure that the received BA frame is not a re-played BA frame from a MITM device. Third, BA frame **502** may include information identifying the TSF of the corresponding soliciting MPDU transmitted by device **400A**. Device **400A** may perform a TSF check on the received BA frame. This may involve validating the received BA frame only if it was transmitted within a predetermined time period (e.g., 256 us) of the TSF of the corresponding soliciting frame **500**. If the TSF of the received BA frame is outside (after) the predetermined time period, device **400A** may discard the received BA frame. As another example, device **400A** may validate the received BA frame only if the hash/TSF of the soliciting frame matches the received BA frame (e.g., with an additional TSF/ID of the soliciting frame to avoid the BA frame transmission as a response to fake frames). Fourth, device **400A** may validate the MIC **520** (FIG. 5) of the received BA frame (e.g., given that the content of the BA cannot be modified by device **402**). If the received BA frame has an invalid MIC **520**, device **400A** may discard the BA frame. Device **400A** may perform one or more of these integrity checks on a received BA frame **502** (e.g., at operation **610** of FIG. 6) before or after beginning to transmit its next frame **500**.

[0134] The data frame integrity checking operations performed by device **400B** (e.g., FIGS. 10-11) may be specified by the communication protocol governing wireless communication link **404** (FIG. 4). The BA frame integrity checking operations performed by device **400A** may also be specified

by the communication protocol (e.g., an IEEE 802.11 protocol). Device **400A** may integrity check a received BA frame before or after TXOP continuation (e.g., a subsequent PPDU transmission). The communication protocol may also define an integrity check mode for BA frame **502**. This may include a STA signaling, for uplink (UL) transmissions, and an AP signaling, for downlink (DL) transmissions: (1) a minimum MPDU start time (e.g., minimum MPDU start spacing **802** of FIG. **8**), (2) a minimum EOF padding value (e.g., minimum EOF padding duration **804** of FIG. **8**), (3) a minimum BA duration (e.g., minimum duration **902** of FIG. **9A**), and/or a (4) minimum BA padding (e.g., for padding **900** of FIGS. **9A** and **9B**). Information identifying one or more of these values may be conveyed (signaled) between devices **400A** and **400B** (e.g., in MAC header **510** or another field of frame **500** and/or in a header, preamble, or other field of BA frame **502**). This signaling may be at the link level, the multi-link device (MLD) level, or the multi-layer multi-domain (MLMD) level.

[0135] FIG. **13** is a diagram showing an exemplary signaling scheme between an AP and a STA when the AP is part of an MLMD AP **1300** and when the STA is part of an MLMD STA (physical device) **1302**. As shown in FIG. **13**, APs (e.g., AP **102** or **104** of FIG. **1**) may exchange signals with corresponding STAs (e.g., STA **106A** or **106B** of FIG. **1**) at the link level and over the air (OTA). Multiple APs may be logically grouped together to form a corresponding MLD level (physical) device. MLMD AP **1300** is a virtual device and may include multiple MLD level devices. The APs of a given MLD level device communicate with a corresponding set of STAs that are logically grouped together at the MLD level. MLMD STA **1302** may include multiple sets of STAs grouped together at the MLD level. A given AP may form one of device **400A** or device **400B** of FIG. **4**, which communicates with a STA that forms the other of device **400A** or **400B**.

[0136] An MLMD STA may be implemented with different radios that are capable of operating at different frequencies. For instance, one radio may operate in a 2.4 GHz band, a second radio may operate in a 5 GHz band, and a third radio may operate in a 6 GHz band. These radios may have different capabilities for the four parameters to configure additional time needed to calculate or verify the integrity of data frames or BA frames. The MLMD level configurations may configure band-specific values for the four signaled values. In this configuration, the band-specific four signaled values are inherited for each link in the specific band.

[0137] As one example, each AP or STA (UL or DL) may configure different ones of the four signaled values (minimum MPDU start time, minimum EOF padding value, minimum BA duration, and minimum BA padding). These values may be signaled in association by using a multi-link (ML) element or an ML link reconfiguration add or modify frame. As another example, an AP and a set of STAs grouped together at the MLD level (UL and DL) may configure the four signaled values. In this example, the values may be kept the same at the MLD AP level. These values may be signaled in (re) association by using an ML element, or in ML link reconfiguration add or modify frames. As a further example, an AP and MLMD STA **1302** may configure the four signaled values. In this example, the values are kept the same within MLMD AP **1300**. The values may be signaled in (re) association using an ML element. A STA or AP may optionally perform header and data payload MIC integrity checks prior to BA frame transmission, even if not required (e.g., the protocol-defined integrity check mode may be a minimum requirement).

[0138] The example of FIG. **5** in which HDR MIC **512** is located between MAC header **510** and the data payload of frame **500** (e.g., MPDU field **514**) is illustrative and non-limiting. FIGS. **14** shows two examples of other possible locations for the HDR MIC **512** in frame **500** (FIG. **5**). Frame **500A** of FIG. **14** shows an example in which HDR MIC **512** is placed between different fields of MAC header **510** (e.g., between a header (HDR) PN field and a Galois counter mode protection (GCMP) HDR field). Frame **500B** of FIG. **14** shows another example in which HDR MIC **512** is placed after the data payload **1402** in frame **500** (e.g., after the payload MIC for data payload **1402**). HDR MIC **512** may be an eight octet field having 64 total bits, for example.

[0139] Frame **500** (e.g., frame **500A** and frame **500B**) may include a data payload MIC **1404**

appended to the end of its data payload **1402** (e.g., where the payload and payload MIC form part of the MPDU field **514** shown in FIG. 5). Payload MIC **1404** may be calculated based on data payload **1402**. On the other hand, HDR MIC **512** is calculated based on the fields of MAC header **510** (e.g., the fields of MAC header **510** not including HDR MIC **512**). Portions **1400** of frames **500A** and **500B** may be portions that are integrity protected by HDR MIC **512**. Data payload **1402** is integrity protected by payload MIC **1404**. The remaining portions of frames **500A** and **500B** may not have integrity protection.

[0140] Device **400A** may separately and independently generate/calculate the HDR MIC and the payload MIC while generating frame **500**. Device **400B** may separately verify the HDR MIC and the payload MIC. Device **400B** may integrity check both header MIC **512** (e.g., at operation **1024** of FIG. 10, operation **1122** of FIG. 11, or operation **1222** of FIG. 12) and payload MIC **1404** (e.g., at operation **1024** of FIG. 10, operation **1126** of FIG. 11, or operation **1224** of FIG. 12).

[0141] If desired, both the payload MIC **1404** (e.g., integrity protection for data payload **1402**) and the HDR MIC **512** may be combined into a single combined MIC field. FIG. 15 shows one example of how frame **500** may include a single MIC that combines the integrity protection of HDR MIC **512** with the integrity protection of payload MIC **1404**. As shown in FIG. 15, frame **500** includes a combined MIC field such as combined MIC **1500**. Device **400A** may generate combined MIC **1500** based on MAC header **510** and data payload **1402**. Device **400A** may, for example, generate HDR MIC **512** (FIG. 14) based on MAC header **510** (e.g., by inputting MAC header **510** to a hashing function), may generate payload MIC **1404** (FIG. 14) based on data payload **1402** (e.g., by inputting data payload **1402** to a hashing function), and by combining HDR MIC **512** with payload MIC **1404** (e.g., using an exclusive OR (XOR) function that outputs the value of HDR MIC **512** as XORED with payload MIC **1404**). Combined MIC **1500** may therefore sometimes also be referred to herein as XOR MIC **1500** or the XOR **1500** of payload MIC **1404** and HDR MIC **512**. In this implementation, device **400B** may verify the integrity of both data payload **1402** and MAC header **510** at the same time (e.g., after BA frame transmission as shown in FIG. 10 or before BA frame transmission as shown in FIG. 12). Re-transmissions of frame may produce a different XOR value in XOR MIC **1500** because the MAC header is updated upon re-transmission.

[0142] In some implementations, an AP may transmit a multi-STA BA frame that carries a BA bitmap to a single STA **106** (pairwise protection) or to multiple STAs **106** (broadcast protection) (e.g., while processing operation **608** of FIG. 6). The multi-STA BA frame may include one or more AID fields. The AID fields may be used to identify the STA which BA bitmap is signaled on the BA frame. FIG. 16 shows an example of an illustrative multi-STA BA frame **1600**.

[0143] As shown in FIG. 16, multi-STA BA frame **1600** may include a frame control field, a duration/ID field, address fields, a BA control field **1610**, a list of per AID fields **1602**, and an FCS field. The list of per AID fields **1602** may, for example, form block ACK **518** of FIG. 5 and may include an AID TID information field, a block ACK starting sequence control field **1606**, and a block ACK bitmap **1608**. AID TID info field **1604** may include an AID11 field associated with block ACK bitmap **1608**, an ACK type field associated with block ACK bitmap **1608**, and a TID field associated with block ACK bitmap **1608**. Block ACK starting sequence control field **1606** may include a fragment number field and a starting SN field.

[0144] The fragment number field may specify the size of the bit map (e.g., 64 bits, 32 bits, etc.). The AID11 fields may be used to identify different information such as the HDR PN, soliciting frame TSF, and HDR MIC **512** of the frame **500** that solicited multi-STA BA frame **1600**. FIG. 17 shows a table **1700** outlining different exemplary AID11 fields that may be included in AID TID info field **1604** of multi-STA BA frame **1600**. The first column of table **1700** lists different AID11 fields, the second column lists the content of the corresponding AID11 field, and the third column lists the meaning of the corresponding AID11 field and its contents. If desired, BA control field **1610** (sometimes also referred to as BAR control field **1610**, e.g., a two-bit field) may be followed by a variable-length BAR information field. If desired, the variable-length BAR information field

may be followed by a control MIC field (e.g., similar to the header MICs described herein, which may be included within list **1602**, which may be represented or identified by list **1602**, or which may replace list **1601**), followed by a variable length padding field, followed by a four-bit FCS field. Frame **1600** need not be a multi-STA BA frame in this implementation (e.g., frame **1600** may be any desired BAR frame in a BlockAckReq format). BA control field **1610** (sometimes also referred to herein as BAR control field **1610**) may include, for example, a one-bit reserved field, followed by a four-bit BAR type field, followed by a one-bit protected control field, followed by a 1 bit Key ID field, followed by a five-bit reserved field, followed by a four-bit TID_INFO field. As another example, BA control field **1610** may include a two-bit reserved field after the Key ID field, followed by a one-bit no memory kept field, followed by a one-bit memory configuration tag field, followed by a one-bit management Ack field, followed by a four-bit TID_INFO field.

[0145] In the example of FIG. **17**, a first AID11 field W is defined to include padding (e.g., padding **900** in FIGS. **9A** and **9B**). This may allow extra time for device **400B** to calculate MIC **520** (FIG. **5**). This field may be MIC protected (e.g., provided as an input to the hashing function when generating MIC **520**). A second AID11 field (e.g., AID11 field “10”) may be defined to include the BA bitmap for a first STA **106** (STA1) operating under a first version of the communication protocol (e.g., an IEEE 802.11bn protocol). A third AID11 field (e.g., AID11 field “11”) may be defined to include the BA bitmap for a second STA **106** (STA2) operating under the first version of the communication protocol.

[0146] A fourth AID11 field X may be defined to include or identify the HDR PN of the corresponding soliciting frame **500** and/or to include or identify the hash (e.g., a hash identifier) and/or TSF of the corresponding soliciting frame **500** (e.g., including two octets for the HDR PN and six octets for the TSF of the soliciting frame). A fifth AID11 field Y may be defined to include or identify the MIC protection and/or check sum of the corresponding frame **500** up to this point (e.g., the BA MIC such as MIC **520** of FIG. **5** for STAs operating under the first version of the communication protocol).

[0147] A sixth AID11 field (e.g., AID11 field “12”) may be defined to include a BA bitmap for a third STA **106** (STA3) operating under a second version of the communication protocol (e.g., a legacy version of the communication protocol) that is older than the first version of the communication protocol. A seventh AID11 field (e.g., AID11 field “13”) may be defined to include a BA bitmap for a fourth STA **106** (STA4) operating under a second version of the communication protocol. These fields may include BA bitmaps for legacy STAs without MIC protection. An eighth AID11 field Z may be defined to include padding (e.g., padding **900** in FIGS. **9A** and **9B**). The example of FIG. **17** is illustrative and non-limiting and, in general, any desired AID11 fields may be specified for any desired purposes.

[0148] FIG. **18** shows two additional examples of frames **500** that may be transmitted by device **400A** as a soliciting frame for a BA frame **502** transmitted by device **400B**. Frame **500C** is one example in which frame **500** is a block acknowledgment request (BAR) frame **500C**. As shown by BAR frame **500C**, information identifying the TSF of frame **500C** may be inserted into HDR PN field **1800** within MAC header **510** (e.g., prior to the payload, which may include a variable length control payload). The same TSF may be present in all HDR PN values. Frame **500D** is another example in which frame **500** is a data or management frame. As shown by data or management frame **500D**, information identifying the TSF of frame **500D** may be inserted into HDR PN field **1800** within MAC header **510**. This may be followed by HDR MIC field **512**, data payload **1402**, payload MIC **1404**, etc.

[0149] The information identifying the TSF of frame **500C** may be inserted into HDR PN field **1700** in any desired manner. As one example, HDR PN field **1800** may include a first numerical value (e.g., one or more HDR_Number bits) followed by a second numerical value (e.g., one or more bits) that identifies the TSF. The first numerical value may be identified using a first octet and a portion of a second octet (e.g., the most significant bit(s) of the second octet), for example. The

second numerical value, identifying the TSF, may be stored in a remainder (e.g., the least significant bit(s)) of the second octet. The first numerical value may, for example, be increased by one for each integrity protection (e.g., up to a PN for 2.sup.10=1,024 MPDUs, corresponding to a maximum BA window size). The TSF may be set to the TSF of the corresponding PPDU transmission time. The TSF may range from a smallest TSF (time) value such as 2.sup.4 us=16 us to a largest TSF (time) value such as 2²¹=1,008 us. The receiver may drop received frames with a TSF difference larger than +/-256 us to its internal TSF value, for example. In a Nonce implementation, the HDR PN field **1800** may be padded with four octets of TSF to make the Nonce **6** octets if desired. As another example, HDR PN field **1800** may include a general number. In Nonce, HDR PN field **1800** is padded with two octets or all ones to make the Nonce six octets in length. This alternative may allow for the replay of frames and does not require TSF synchronization.

[0150] Additionally or alternatively, frame **500** may identify that it is a soliciting frame by calculating a hash on the soliciting frame. In these implementations, device **400A** may identify that frame **500** is a soliciting frame using frame control, duration, retry, preamble, and addresses (e.g., constant values not depending on receive operation). This generally does not require a TSF-based nonce but may require collection of a large amount of data. Alternatively, device **400A** may identify that frame **500** is a soliciting frame using frame control, duration, retry, preamble, addresses, and smallest received sequence number (SN) and HDR PN value techniques. This information is based on BA frame content and may add processing time and complexity.

[0151] Consider an example in which an AP and STA have set up a target wakeup time (TWT) service period (SP) and in which an attacker (e.g., device **402** of FIG. 4) receives the AP transmissions but causes interference to the STA that prevents the STA from receiving frames. In the next TWT SP, the attacker might reply with a frame that will early terminate the TWT SP. The STA may be confused and not receive any frames in the TWT SP. The attacker may block Buffer Status information transmission and may retransmit the old Buffer Status information at a later time. This may cause incorrect allocations for the STA. This attack scenario leads to two new protections for the transmitted frames.

[0152] First, MIC protection of the MPDU header, MPDU data payload, and BA frames may be used to avoid modifications and fake frames. MIC protection generally offers poor protection against replays of non-received frames. MIC protection may also add delays if performed prior to transmission of the BA frame. Second, TSF may be utilized to avoid replays of old data and BA frames. Replay protection may be irrelevant for MAC header information because the header information may change over time. However, TSF-based protection may protect against replays of MPDUs by the attacker. BA frames may include replay protection that ensures that the BA frame **502** is transmitted as a response to a corresponding soliciting frame **500**. BA frame replay protection is especially relevant when the data receiver performs integrity checks after transmission of the BA frame.

[0153] FIG. 19 shows an example in which an MLD transmitter **1900** transmits frames **500** to an MLD receiver **1902** and in which MLD receiver **1902** transmits BA frames **502** to MLD transmitter **1900**. MLD transmitter **1900** may include a set of STAs including STA A1, STA A2, and STA A3 (e.g., AP STAs or non-AP STAs). MLD receiver **1902** may include a set of STAs include STA B1, STA B2, and STA B3 (e.g., non-AP STAs or AP STAs).

[0154] As shown in FIG. 19, MLD transmitter **1900** controls the MPDU sequence numbers that are transmitted to MLD receiver **1902** (e.g., MLD transmitter **1900** may control SN assignment, PN assignment, single TX window assignment, and ML TX/Rc-TX management). Transmitter **1900** has a transmit buffer **1904** that stores all frames in transmission (see, e.g., transmit buffer **406** of FIG. 4). Frames that are currently in transmission are stored in STA-specific buffers (see, e.g., temporary buffer **408** of FIG. 4). MLD receiver **1902** receives frames **500**, temporarily stores information from the frames in scoreboards for different STAs, and then provides information from

the scoreboards to a single reordering buffer that performs duplicate detection, PN checking, etc. [0155] MLD receiver **1902** may acknowledge all received frames **500** but has no control over the MPDUs in transmission. If a stateless BA is assumed, BA bitmaps may signal, for each MPDU that is acknowledged in a BA frame: that an MPDU is received (e.g., using a value “1”) or that an MPDU is not received or that the receiver otherwise has no knowledge of the MPDU (e.g., using a value “0”). When the BA frame is integrity protected, MLD transmitter **1900** needs to integrity check the BA frame before updating its transmission buffer. This may eliminate attacks using fake BA frames.

[0156] Consider another example in which device **400B** checks both the HDR MIC and the payload MIC of a received frame **500** after transmitting BA frame **502**. If care is not taken, this can produce a BA off-synchronization problem. In this example, device **400B** may end up discarding a received frame after transmitting the corresponding BA frame **502**. This can produce a hole in the reorder buffer on device **400B**. This data is not forwarded to its corresponding application until the hole is filled or the BA window moves forward. However, the BA frame signals to device **400A** that the transmission of frame **500** was acceptable to device **400B**. Device **400A** can then erase the frame from its transmission buffer and does not know that device **400B** is waiting for retransmission of the frame.

[0157] In a first example, this BA off-synchronization problem may be solved using a silent reorder buffer update at device **400B**. In this example, device **400B** may detect that a payload MIC or a header MIC of the received frame **500** fails for MPDUs in the received frame after transmitting the corresponding BA frame. However, device **400B** may include internal logic that checks when to forward the MPDUs to its reorder buffer. Device **400B** may, for example, check whether device **400A** will have enough time to retransmit the MPDUs. Device **400B** may, for example, check whether device **400A** has progressed on transmitted SN values and has sent larger SNs than already received. If this internal condition is met, device **400B** may silently forward the MPDUs to its reorder buffer. This will cause loss of MPDUs having invalid integrity after transmission of the BA frame.

[0158] An example of this type of solution is shown in the timing diagram of FIG. **20**. In the example of FIG. **20**, device **400A** transmits MPDUs of SN=11 and 12 at time T1. Device **400B** may store a previously received and integrity checked MPDU of SN=13 in its reorder buffer but has not yet integrity checked the received MPDUs of SN=11 and 12, so the received MPDUs of SN=11 and 12 have not yet been passed to the reorder buffer. Device **400B** transmits a BA frame **502** acknowledging the MPDUs of SN=11 and 12 at time T2, despite having not yet integrity checked the MPDUs of SN=11 and 12.

[0159] At time T3, device **400B** has finished calculating that the received MPDUs of SN=11 and 12 have failed the integrity check. Device **400B** may then discard the invalid MPDUs of SN=11 and 12. However, since device **400B** has already transmitted the BA frame for MPDUs of SN=11 and 12, device **400A** incorrectly believes that device **400B** has received the MPDUs of SN=11 and 12, causing device **400A** to delete the MPDUs of SN=11 and 12 from its transmission buffer.

[0160] Device **400A** may then transmit the next MPDU having SN=14 to device **400B**. Device **400B** receives the MPDU of SN=14, transmits a corresponding BA frame for the MPDU of SN=14 to device **400A**, and then determines at time T4 that the MPDU of SN=14 has passed the integrity check. Upon passing the integrity check, device **400B** places the MPDU of SN=14 into the reorder buffer (e.g., in the correct sequential order after the MPDU of SN=13). The internal logic on device **400B** may silently forward the MPDU of SN=14 to the reorder buffer while ignoring the discarded MPDUs of SN=11 and 12, causing loss of those MPDUs but re-synchronizing device **400B** to device **400A** (e.g., solving the BA off-synchronization problem).

[0161] In another example, shown in the timing diagram of FIG. **21**, the BA off-synchronization problem may be solved using a delay to the transmit buffer. As shown in FIG. **21**, rather than waiting for device **400A** to transmit the next MPDU in the sequence after the MPDUs of SN=11

and 12 fail the MIC check at time T3, device **400B** may instead transmit an integrity-protected BAR frame to device **400A** that identifies the dropped MPDUs of SN=11 and 12. This signaling of the dropped MPDUs or frames is also referred to as reorder buffer status signaling. The BAR frame may signal or identify holes in the reorder buffer on device **400B** (e.g., as caused by MIC check failures or holes that cause delays to low delay frames forwarding to the corresponding application).

[0162] At time TB, device **400A** may transmit a BA frame in response to the BAR frame received from device **400B**. The transmitter may apply a transmission buffer update delay (e.g., “TransmissionBufferUpdateDelay”). This delay may allow sufficient time for device **400A** to re-transmit the MPDUs of SN=11 and 12. Device **400B** may transmit a BA frame in response to the re-transmission of the MPDUs of SN=11 and 12 and may then determine that the re-transmitted MPDUs have valid integrity. Device **400A** may then transmit its next MPDU of SN=14 to device **400B**.

[0163] As another example, if desired, device **400B** may transmit a multi-STA frame **1600** (FIG. 16) to signal to device **400A** that its reorder buffer is missing MPDUs transmitted by device **400A** (e.g., at time TA of FIG. 21). In this example, the BA control field **1610** of multi-STA BA frame **1600** (FIG. 16) may include information identifying the status of the reorder buffer **412** on device **400B**.

[0164] FIG. 22 shows an example of information that may be stored in the BA control field **1610** of multi-STA BA frame **1600** to signal the reorder buffer status of device **400B** to device **400A**. As shown in FIG. 22, BA control field **1610** may include information about the status of its reorder buffer **412** such as reorder buffer status field **2200**. Reorder buffer status field **2200** may include one or more bits. Reorder buffer status field **2200** may be a first value (e.g., a bit set to “1”) to identify, to device **400A**, that multi-STA BA frame **1600** includes one or more ReorderBufferStatus per AID Fields. Reorder buffer status **2200** may be a second value (e.g., a bit set to “0”) to identify, to device **400A**, that multi-STA BA frame **1600** does not include any ReorderBufferStatus per AID Field. If desired, a new AID field (e.g., an AID11 field) may be defined that signals the ReorderBufferStatus start.

[0165] FIG. 23 shows a table **2300** outlining additional exemplary AID11 fields that may be included in AID TID info field **1604** of multi-STA BA frame **1600** (e.g., a multi-STA BA frame **1600** having reorder buffer status **2200** in its BA control field **1610** as shown in FIG. 22). In the example of FIG. 23, an AID11 field A is reserved to identify the status of the reorder buffer **412** on device **400B**. An AID11 field B may be defined to include a reorder bitmap for the corresponding STA (e.g., STA2) operating under the first version of the communications protocol (e.g., IEEE 802.11bn). This bitmap identifies MPDUs that were discarded by device **400A** after transmission of BA frame **502** (e.g., the MPDUs of SN=11 and 12 in the example of FIG. 21). If desired, an AID11 field (e.g., AID11 field “100”) may be defined to include a BA bitmap for a legacy STA (e.g., STA3) without MIC protection. The example of FIG. 23 is illustrative and non-limiting and, in general, any desired AID11 fields may be specified for any desired purposes.

[0166] In this example, the transmitter (e.g., device **400A**) controls retransmissions and selects a suitable operation after receiving the reorder buffer status from device **400B** (e.g., in multi-STA BA frame **1600**). In a best case, the transmitter may then retransmit a failed MPDU. In a default cause, the transmitter may move the reorder buffer forward. This may re-synchronize the transmitter and receiver block ACKs, may allow blocked data in the reorder buffer to be moved to the corresponding application at the receiver (e.g., device **400B**), and the hole/missed frame may be retransmitted in a TCP retransmission or another higher layer retransmission mechanism. As BA windows get larger (e.g., supporting up to 1024 MPDUs), a hole may block frames for increasingly long times, causing higher transmission delays.

[0167] As used herein, the term “concurrent” means at least partially overlapping in time. In other words, first and second events are referred to herein as being “concurrent” with each other if at

least some of the first event occurs at the same time as at least some of the second event (e.g., if at least some of the first event occurs during, while, or when at least some of the second event occurs). First and second events can be concurrent if the first and second events are simultaneous (e.g., if the entire duration of the first event overlaps the entire duration of the second event in time) but can also be concurrent if the first and second events are non-simultaneous (e.g., if the first event starts before or after the start of the second event, if the first event ends before or after the end of the second event, or if the first and second events are partially non-overlapping in time). As used herein, the term “while” is synonymous with “concurrent.” The term “when” also implies at least some concurrency (e.g., event A occurring “when” event B occurs means that at least some of event A is concurrent with at least some of event B).

[0168] STAs **106** and APs **102/104** (FIG. 1) may gather and/or use personally identifiable information. It is well understood that the use of personally identifiable information should follow privacy policies and practices that are generally recognized as meeting or exceeding industry or governmental requirements for maintaining the privacy of users. In particular, personally identifiable information data should be managed and handled so as to minimize risks of unintentional or unauthorized access or use, and the nature of authorized use should be clearly indicated to users.

[0169] The methods and operations described above in connection with FIGS. **1-23** may be performed by the components of a STA and/or AP using software, firmware, and/or hardware (e.g., dedicated circuitry or hardware). Software code for performing these operations may be stored on non-transitory computer readable storage media (e.g., tangible computer readable storage media) stored on one or more of the components of the STA and/or AP. The software code may sometimes be referred to as software, data, instructions, program instructions, or code. The non-transitory computer readable storage media may include drives, non-volatile memory such as non-volatile random-access memory (NVRAM), removable flash drives or other removable media, other types of random-access memory, etc. Software stored on the non-transitory computer readable storage media may be executed by processing circuitry on one or more of the components of the STA and/or AP. The processing circuitry may include microprocessors, central processing units (CPUs), application-specific integrated circuits with processing circuitry, or other processing circuitry.

[0170] For one or more aspects, at least one of the components set forth in one or more of the preceding figures may be configured to perform one or more operations, techniques, processes, or methods as set forth in the example section below. For example, circuitry associated with an electronic device, authentication server, one or more processors, etc. as described above in connection with one or more of the preceding figures may be configured to operate in accordance with one or more of the examples set forth below in the example section.

EXAMPLES

[0171] In the following sections, further exemplary aspects are provided.

[0172] Example 1 includes a method of operating a first electronic device to wirelessly communicate with a second electronic device, the method comprising: transmitting, using one or more antennas, an integrity-protected frame for receipt by the second electronic device, wherein the integrity-protected frame includes a preamble, a media access control (MAC) header, a delimiter between the preamble and the MAC header, a message integrity check (MIC) field based on the MAC header, a data payload, the MAC header being between the data payload and the delimiter, and a padding field configured to accommodate a processing delay associated with generating or verifying the MIC field.

[0173] Example 2 includes the method of example 1 or some other example or combination of examples herein, wherein the padding field is between the delimiter and the MAC header in the integrity-protected frame.

[0174] Example 3 includes the method of any one of examples 1 or 2 or some other example or combination of examples herein, wherein the padding field comprises a duplicate of the delimiter.

[0175] Example 4 includes the method of any one of examples 1-3 or some other example or combination of examples herein, wherein the delimiter comprises an aggregated MAC protocol data unit (A-MPDU) associated with the data payload.

[0176] Example 5 includes the method of any one of examples 1-4 or some other example or combination of examples herein, wherein the padding field is between the MAC header and the MIC field in the integrity-protected frame.

[0177] Example 6 includes the method of any one of examples 1-5 or some other example or combination of examples herein, wherein the padding field is between the MIC field and the data payload of the integrity-protected frame.

[0178] Example 7 includes the method of any one of examples 1-6 or some other example or combination of examples herein, wherein the padding field is appended to an end of the data payload.

[0179] Example 8 includes the method of any one of examples 1-7 or some other example or combination of examples herein, wherein the integrity-protected frame further comprises: a first MAC protocol data unit (MPDU) field that includes the data payload; a second MPDU field that includes an additional data payload; a first set of padding fields that includes the padding field, the padding fields in the first set comprising duplicates of the delimiter; and a second set of padding fields appended to an end of the second MPDU, the padding fields in the second set comprising end of frame (EOF) bits.

[0180] Example 9 includes the method of any one of examples 1-8 or some other example or combination of examples herein, further comprising: receiving, using the one or more antennas, an integrity-protected block acknowledgment (BA) frame from the second electronic device; and verifying, using one or more processors, an additional MIC field of the BA frame.

[0181] Example 10 includes the method of any one of examples 1-9 or some other example or combination of examples herein, wherein the MAC header comprises a packet number (PN) field and wherein the PN field identifies a first time synchronization function (TSF) of the integrity-protected frame.

[0182] Example 11 includes the method of any one of examples 1-10 or some other example or combination of examples herein, wherein the BA frame identifies a second TSF, the method further comprising: discarding the BA frame when the second TSF is after a predetermined time range from the first TSF.

[0183] Example 12 includes the method of any one of examples 1-11 or some other example or combination of examples herein, wherein the BA frame comprises a hash identifier associated with the integrity-protected frame.

[0184] Example 13 includes the method of any one of examples 1-12 or some other example or combination of examples herein, wherein the integrity-protected frame further comprises: an additional MIC field based on the data payload, the data payload being between the additional MIC field and the MAC header in the integrity-protected frame.

[0185] Example 14 includes the method of any one of examples 1-13 or some other example or combination of examples herein, wherein the MIC field is between the data payload and a packet number (PN) field of the MAC header in the integrity-protected frame.

[0186] Example 15 includes the method of any one of examples 1-14 or some other example or combination of examples herein, wherein the additional MIC field is between the MIC field and the data payload in the integrity-protected frame.

[0187] Example 16 includes the method of any one of examples 1-15 or some other example or combination of examples herein, further comprising: generating, using one or more processors, a first MIC value based on the MAC header; generating, using the one or more processors, a second MIC value based on the data payload; and generating, using the one or more processors, the MIC field by performing an exclusive OR (XOR) operation on the first MIC value and the second MIC value, the data payload being between the MIC field and the MAC header in the integrity-protected

frame.

[0188] Example 17 includes a method of operating a first electronic device to wirelessly communicate with a second electronic device, the method comprising: receiving, using one or more antennas, a frame; and transmitting, using the one or more antennas, an integrity-protected block acknowledgement (BA) frame for receipt by the second electronic device, wherein the integrity-protected BA frame includes a preamble, a block acknowledgment (ACK) to a media access control protocol data unit (MPDU) in the frame, a message integrity check (MIC) field based on the block ACK, the block ACK being between the MIC field and the preamble in the integrity-protected BA frame, and a padding field configured to accommodate a processing delay associated with generating or verifying the MIC field.

[0189] Example 18 includes the method of example 17 or some other example or combination of examples herein, wherein the padding field is between the block ACK and the MIC field in the integrity-protected BA frame.

[0190] Example 19 includes the method of any one of examples 17, 18, or some other example or combination of examples herein, wherein the padding field is between the block ACK and the preamble in the integrity-protected BA frame.

[0191] Example 20 includes the method of any one of examples 17-19 or some other example or combination of examples herein, wherein the padding field comprises an association identifier (AID) traffic identifier (TID) Info field that is unassociated with any station (STA) in communication with the first and second electronic devices.

[0192] Example 21 includes the method of any one of examples 17-20 or some other example or combination of examples herein, wherein the integrity-protected BA frame comprises a block ACK bitmap and the padding field comprises an association identifier 11 (AID11) field associated with the block ACK bitmap.

[0193] Example 22 includes the method of any one of examples 17-21 or some other example or combination of examples herein, wherein the integrity-protected BA frame comprises a BA control field and the BA control field includes one or more bits identifying a status of a reorder buffer on the first electronic device.

[0194] Example 23 includes the method of any one of examples 17-22 or some other example or combination of examples herein, wherein the integrity-protected BA frame comprises an association identifier 11 (AID11) field that identifies one or more MPDUs that were discarded from a reorder buffer on the first electronic device.

[0195] Example 24 includes a method of operating a first electronic device to wirelessly communicate with a second electronic device, the method comprising: receiving, using one or more antennas, a frame; transmitting, using the one or more antennas, a block acknowledgement (BA) frame for receipt by the second electronic device, wherein the BA frame includes a block acknowledgment (ACK) to a media access control protocol data unit (MPDU) in the frame; and attempting to verify, using one or more processors prior to transmission of the integrity-protected BA frame, integrity of a media access control (MAC) header of the MPDU in the frame.

[0196] Example 25 includes the method of example 24 or some other example or combination of examples herein, further comprising: attempting to verify, using the one or more processors after transmission of the integrity-protected BA frame, integrity of a data payload of the MPDU in the frame.

[0197] Example 26 includes the method of any one of example 24, 25, or some other example or combination of examples herein, further comprising: attempting to verify, using the one or more processors prior to transmission of the integrity-protected BA frame, integrity of a data payload of the MPDU in the frame.

[0198] Example 27 includes a method of operating a first electronic device to wirelessly communicate with a second electronic device, the method comprising: receiving, using one or more antennas, a frame; transmitting, using the one or more antennas, a block acknowledgement (BA)

frame for receipt by the second electronic device, wherein the BA frame includes a block acknowledgment (ACK) to a media access control protocol data unit (MPDU) in the frame; calculating, using one or more processors after of the integrity-protected BA frame, a message integrity check (MIC) value based on at least some of the MPDU; and transmitting, using the one or more antennas, a block acknowledgment request (BAR) frame when the calculated MIC value does not match a corresponding MIC field included in the frame, wherein the BAR frame identifies the MPDU.

[0199] Example 28 may include an apparatus comprising means to perform one or more elements of a method described in or related to any of examples 1-27 or any combination thereof, or any other method or process described herein.

[0200] Example 29 may include one or more non-transitory computer-readable media comprising instructions to cause an electronic device, upon execution of the instructions by one or more processors of the electronic device, to perform one or more elements of a method described in or related to any of examples 1-27 or any combination thereof, or any other method or process described herein.

[0201] Example 30 may include an apparatus comprising logic, modules, or circuitry to perform one or more elements of a method described in or related to any of examples 1-27 or any combination thereof, or any other method or process described herein.

[0202] Example 31 may include a method, technique, or process as described in or related to any of examples 1-27 or any combination thereof, or portions or parts thereof.

[0203] Example 32 may include an apparatus comprising: one or more processors and one or more non-transitory computer-readable storage media comprising instructions that, when executed by the one or more processors, cause the one or more processors to perform the method, techniques, or process as described in or related to any of examples 1-27, or any combination thereof, or portions thereof.

[0204] Example 33 may include a signal as described in or related to any of examples 1-27, or any combination thereof, or portions or parts thereof.

[0205] Example 34 may include a datagram, information element, packet, frame, segment, PDU, or message as described in or related to any of examples 1-27, or any combination thereof, or portions or parts thereof, or otherwise described in the present disclosure.

[0206] Example 35 may include a signal encoded with data as described in or related to any of examples 1-26, or any combination thereof, or portions or parts thereof, or otherwise described in the present disclosure.

[0207] Example 36 may include a signal encoded with a datagram, IE, packet, frame, segment, PDU, or message as described in or related to any of examples 1-26, or any combination thereof, or portions or parts thereof, or otherwise described in the present disclosure.

[0208] Example 37 may include an electromagnetic signal carrying computer-readable instructions, wherein execution of the computer-readable instructions by one or more processors is to cause the one or more processors to perform the method, techniques, or process as described in or related to any of examples 1-27, or any combination thereof, or portions thereof.

[0209] Example 38 may include a computer program comprising instructions, wherein execution of the program by a processing element is to cause the processing element to carry out the method, techniques, or process as described in or related to any of examples 1-27, or any combination thereof, or portions thereof.

[0210] Example 39 may include a signal in a wireless network as shown and described herein.

[0211] Example 40 may include a method of communicating in a wireless network as shown and described herein.

[0212] Example 41 may include a system for providing wireless communication as shown and described herein.

[0213] Example 42 may include a device for providing wireless communication as shown and

described herein.

[0214] Any of the above-described examples may be combined with any other example (or combination of examples), unless explicitly stated otherwise. The foregoing description of one or more implementations provides illustration and description but is not intended to be exhaustive or to limit the scope of aspects to the precise form disclosed.

Claims

1. A method of operating a first electronic device to wirelessly communicate with a second electronic device, the method comprising: transmitting, using one or more antennas, an integrity-protected frame for receipt by the second electronic device, wherein the integrity-protected frame includes a preamble, a media access control (MAC) header, a delimiter between the preamble and the MAC header, a message integrity check (MIC) field based on the MAC header, a data payload, the MAC header being between the data payload and the delimiter, and a padding field configured to accommodate a processing delay associated with generating or verifying the MIC field.
2. The method of claim 1, wherein the padding field is between the delimiter and the MAC header in the integrity-protected frame.
3. The method of claim 2, wherein the padding field comprises a duplicate of the delimiter and wherein the delimiter comprises an aggregated MAC protocol data unit (A-MPDU) associated with the data payload.
4. The method of claim 1, wherein the padding field is between the MAC header and the MIC field in the integrity-protected frame.
5. The method of claim 1, wherein the padding field is between the MIC field and the data payload of the integrity-protected frame.
6. The method of claim 1, wherein the padding field is appended to an end of the data payload.
7. The method of claim 1, wherein the integrity-protected frame further comprises: a first MAC protocol data unit (MPDU) field that includes the data payload; a second MPDU field that includes an additional data payload; a first set of padding fields that includes the padding field, the padding fields in the first set comprising duplicates of the delimiter; and a second set of padding fields appended to an end of the second MPDU, the padding fields in the second set comprising end of frame (EOF) bits.
8. The method of claim 1, further comprising: receiving, using the one or more antennas, a block acknowledgment (BA) frame from the second electronic device; and verifying, using one or more processors, an additional MIC field of the BA frame.
9. The method of claim 8, wherein the MAC header comprises a packet number (PN) field and wherein the PN field identifies a first time synchronization function (TSF) of the integrity-protected frame and wherein the BA frame identifies a second TSF, the method further comprising: discarding the BA frame when the second TSF is after a predetermined time range from the first TSF.
10. The method of claim 8, wherein the BA frame comprises a hash identifier associated with the integrity-protected frame.
11. The method of claim 1, wherein the integrity-protected frame further comprises: an additional MIC field based on the data payload, the data payload being between the additional MIC field and the MAC header in the integrity-protected frame.
12. The method of claim 11, wherein the MIC field is between the data payload and a packet number (PN) field of the MAC header in the integrity-protected frame.
13. The method of claim 11, wherein the additional MIC field is between the MIC field and the data payload in the integrity-protected frame.
14. The method of claim 1, further comprising: generating, using one or more processors, a first MIC value based on the MAC header; generating, using the one or more processors, a second MIC

value based on the data payload; and generating, using the one or more processors, the MIC field by performing an exclusive OR (XOR) operation on the first MIC value and the second MIC value, the data payload being between the MIC field and the MAC header in the integrity-protected frame.

15. A method of operating a first electronic device to wirelessly communicate with a second electronic device, the method comprising: receiving, using one or more antennas, a frame; and transmitting, using the one or more antennas, an integrity-protected block acknowledgement (BA) frame for receipt by the second electronic device, wherein the integrity-protected BA frame includes a preamble, a block acknowledgment (ACK) to a media access control protocol data unit (MPDU) in the frame, a message integrity check (MIC) field based on the block ACK, the block ACK being between the MIC field and the preamble in the integrity-protected BA frame, and a padding field configured to accommodate a processing delay associated with generating or verifying the MIC field.

16. The method of claim 15, wherein the padding field is between an end of the MIC field and a beginning of a frequency checksum field of the integrity-protected BA frame.

17. The method of claim 15, wherein the padding field is between the block ACK and the MIC field in the integrity-protected BA frame.

18. A method of operating a first electronic device to wirelessly communicate with a second electronic device, the method comprising: receiving, using one or more antennas, a frame; transmitting, using the one or more antennas, a block acknowledgement (BA) frame for receipt by the second electronic device, wherein the BA frame includes a block acknowledgment (ACK) to a media access control protocol data unit (MPDU) in the frame; and attempting to verify, using one or more processors prior to transmission of the integrity-protected BA frame, integrity of a media access control (MAC) header of the MPDU in the frame.

19. The method of claim 18, further comprising: attempting to verify, using the one or more processors after transmission of the integrity-protected BA frame, integrity of a data payload of the MPDU in the frame.

20. The method of claim 18, further comprising: attempting to verify, using the one or more processors prior to transmission of the integrity-protected BA frame, integrity of a data payload of the MPDU in the frame.
